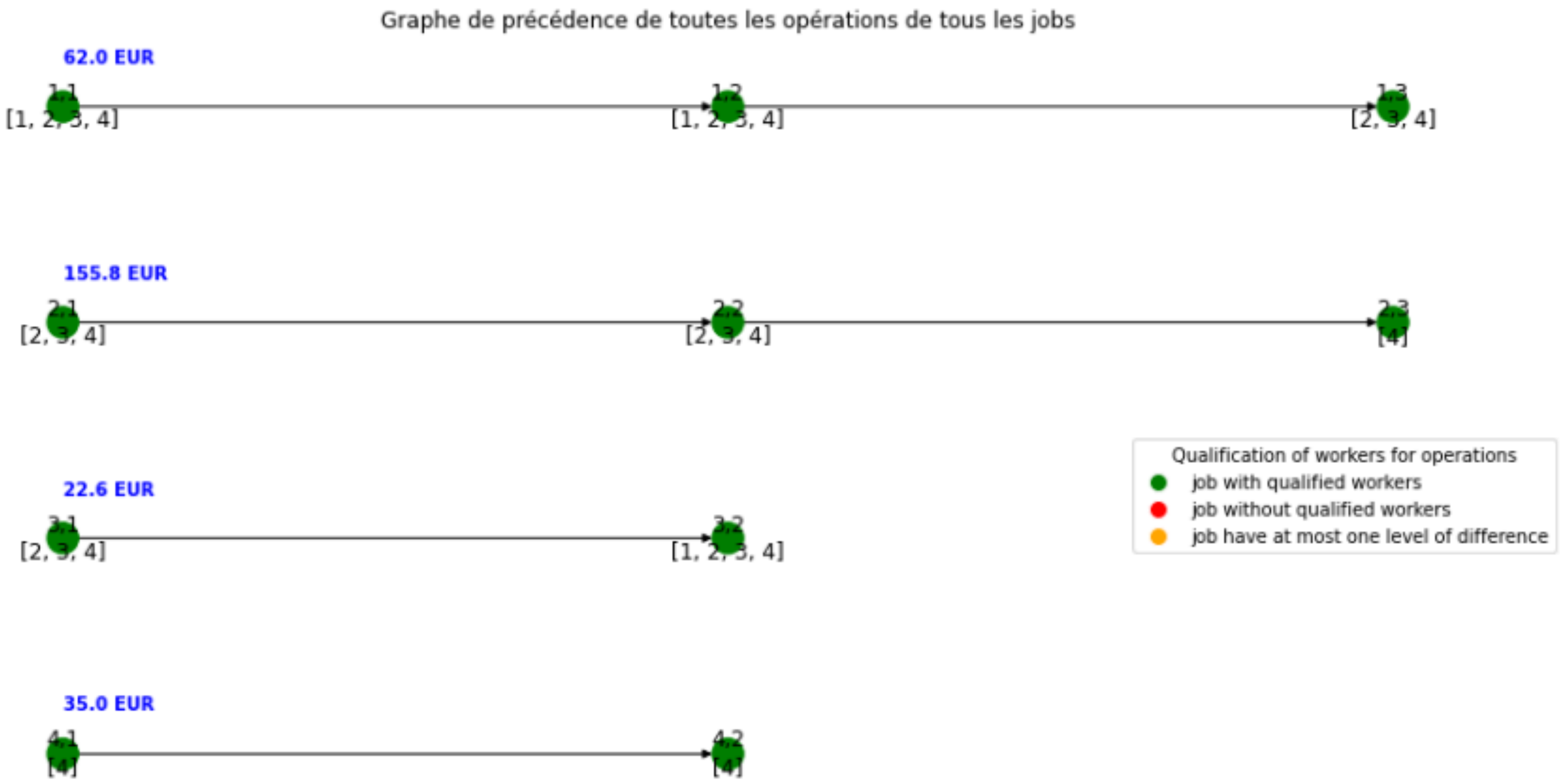
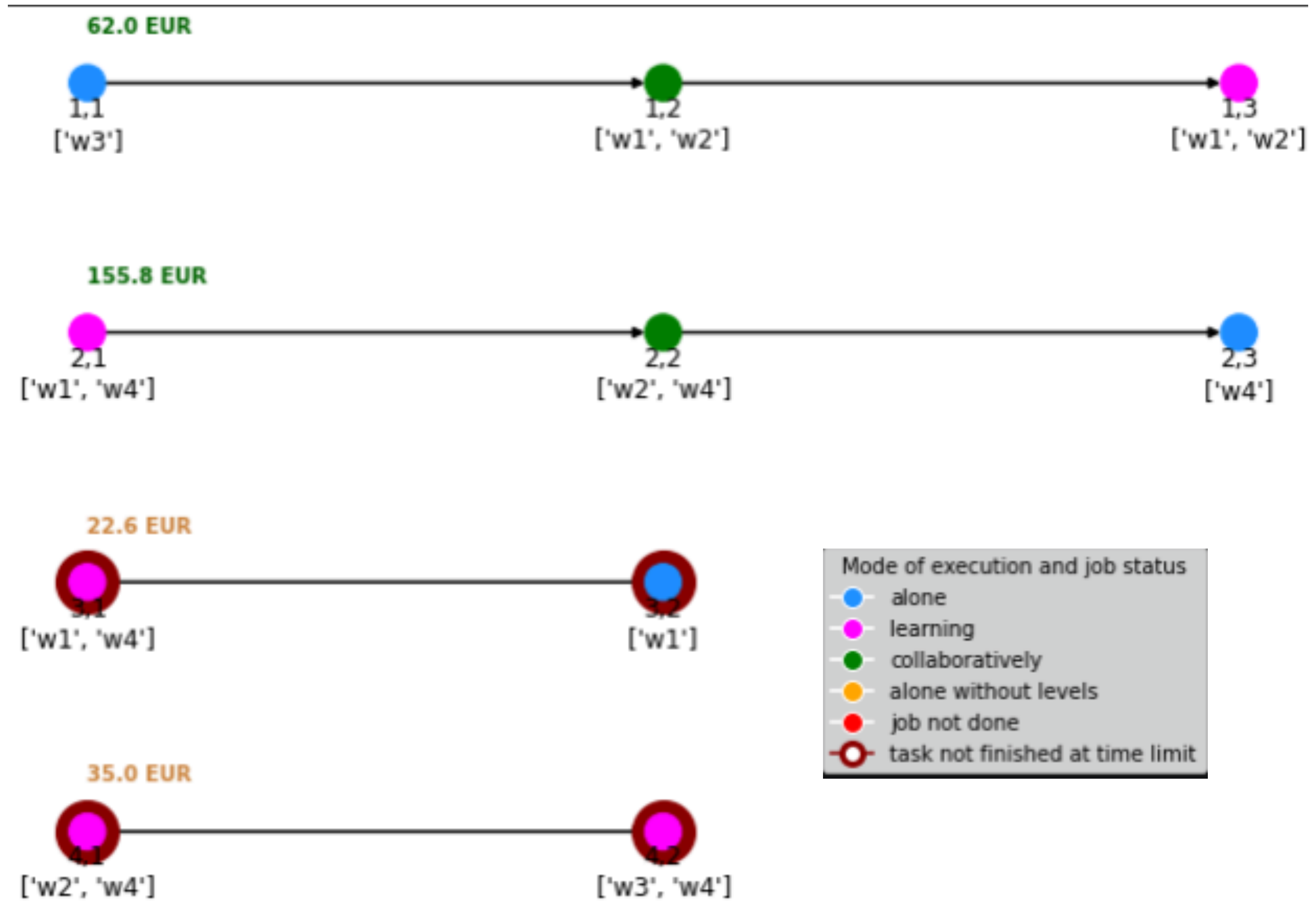
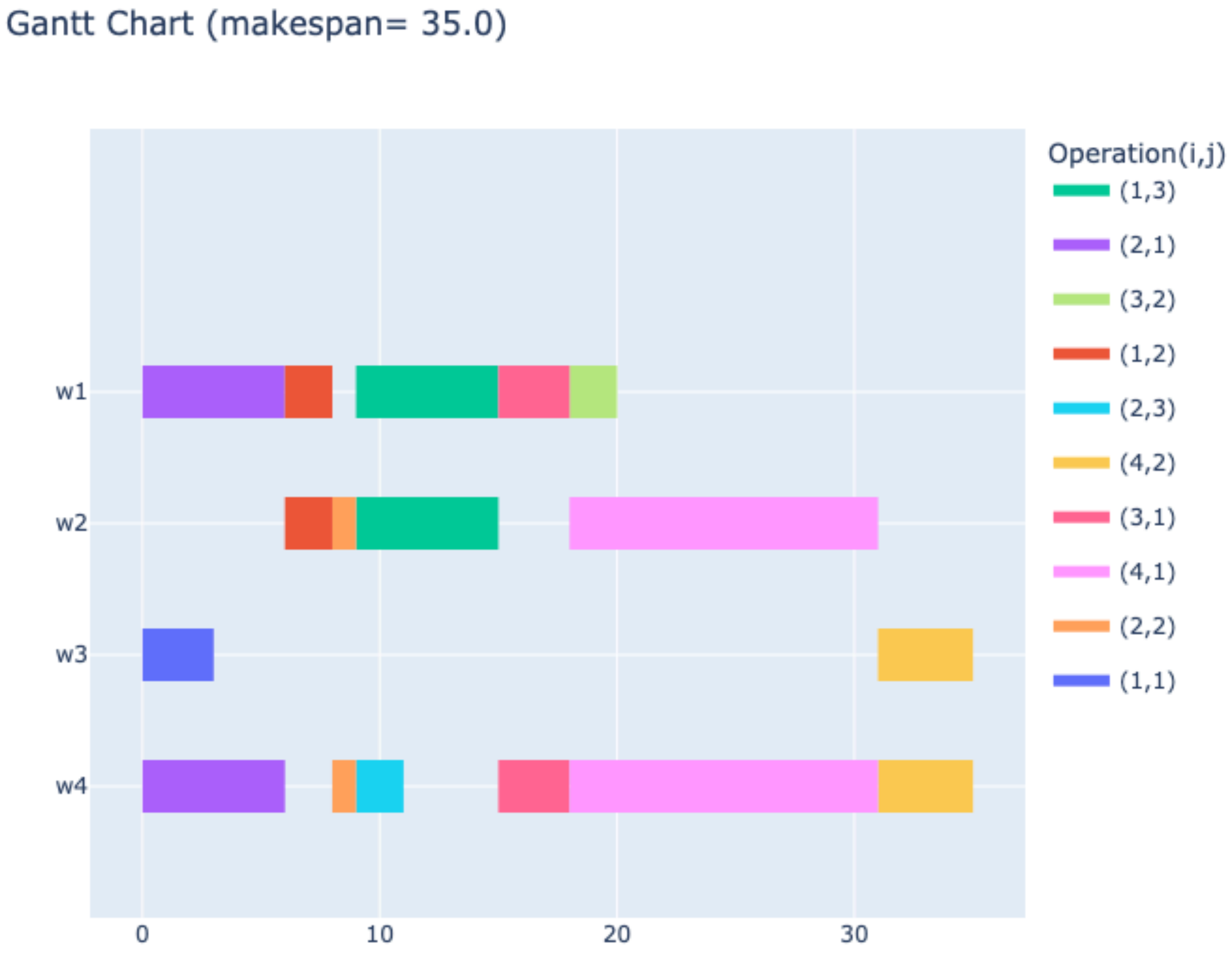
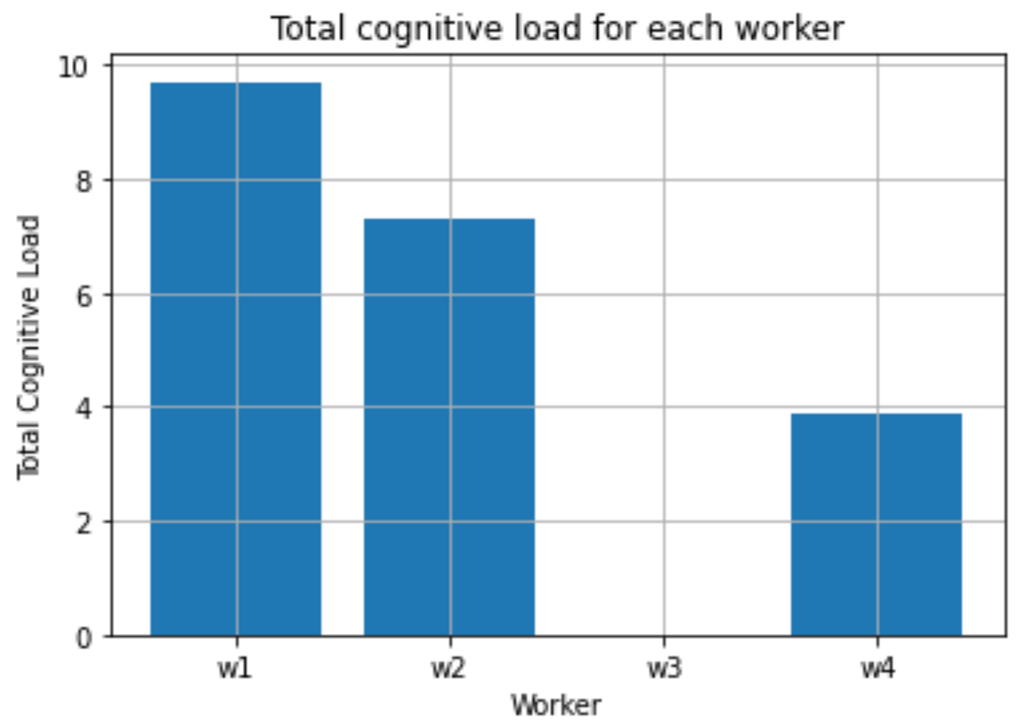

Graphe de précédence de toutes les opérations de tous les jobs
62.0 EUR
1,1
[1, 2, 3, 4]
1,2
[1, 2, 3, 4]
1,3
[2, 3, 4]
155.8 EUR
2,1
[2, 3, 4]
2,2
[2, 3, 4]
2,3
[4]
22.6 EUR
3,1
[2, 3, 4]
3,2
[1, 2, 3, 4]
35.0 EUR
4,1
[4]
4,2
[4]
Qualification of workers for operations
job with qualified workers
job without qualified workers
job have at most one level of difference

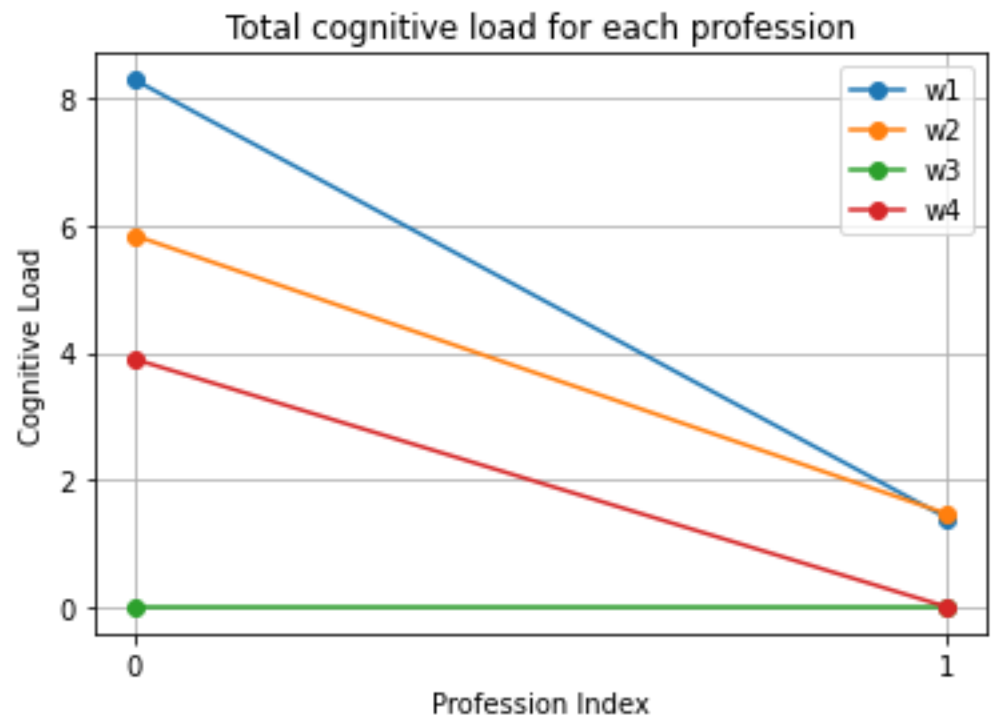
Random Search

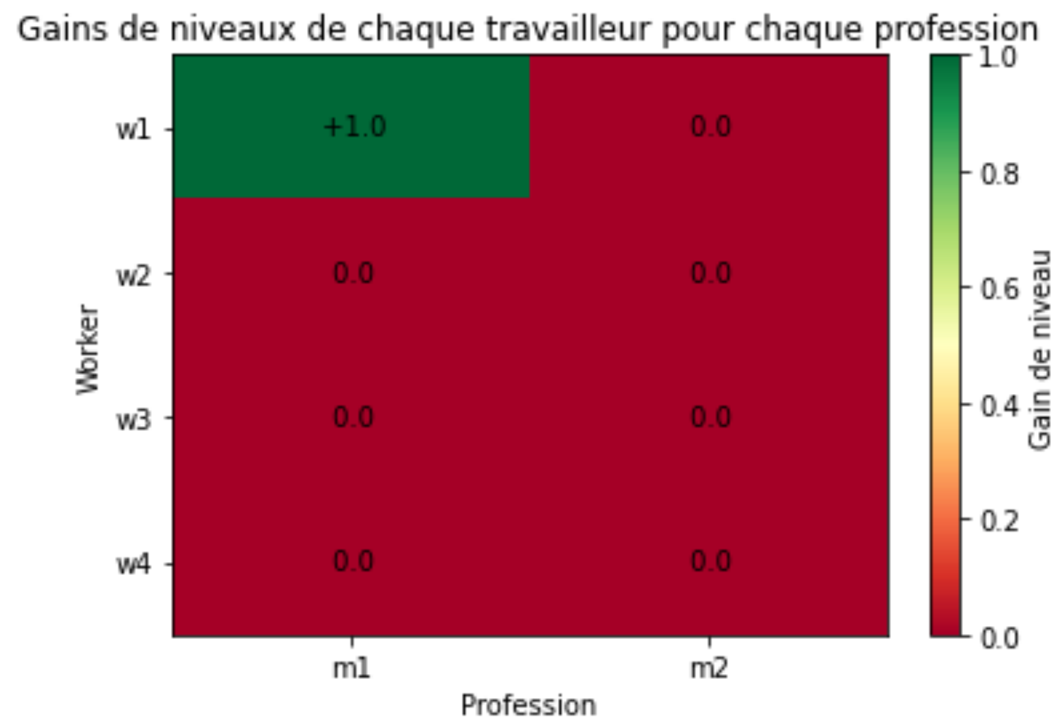
Génere plusieurs Solutions aléatoire et séléctionne la meilleure





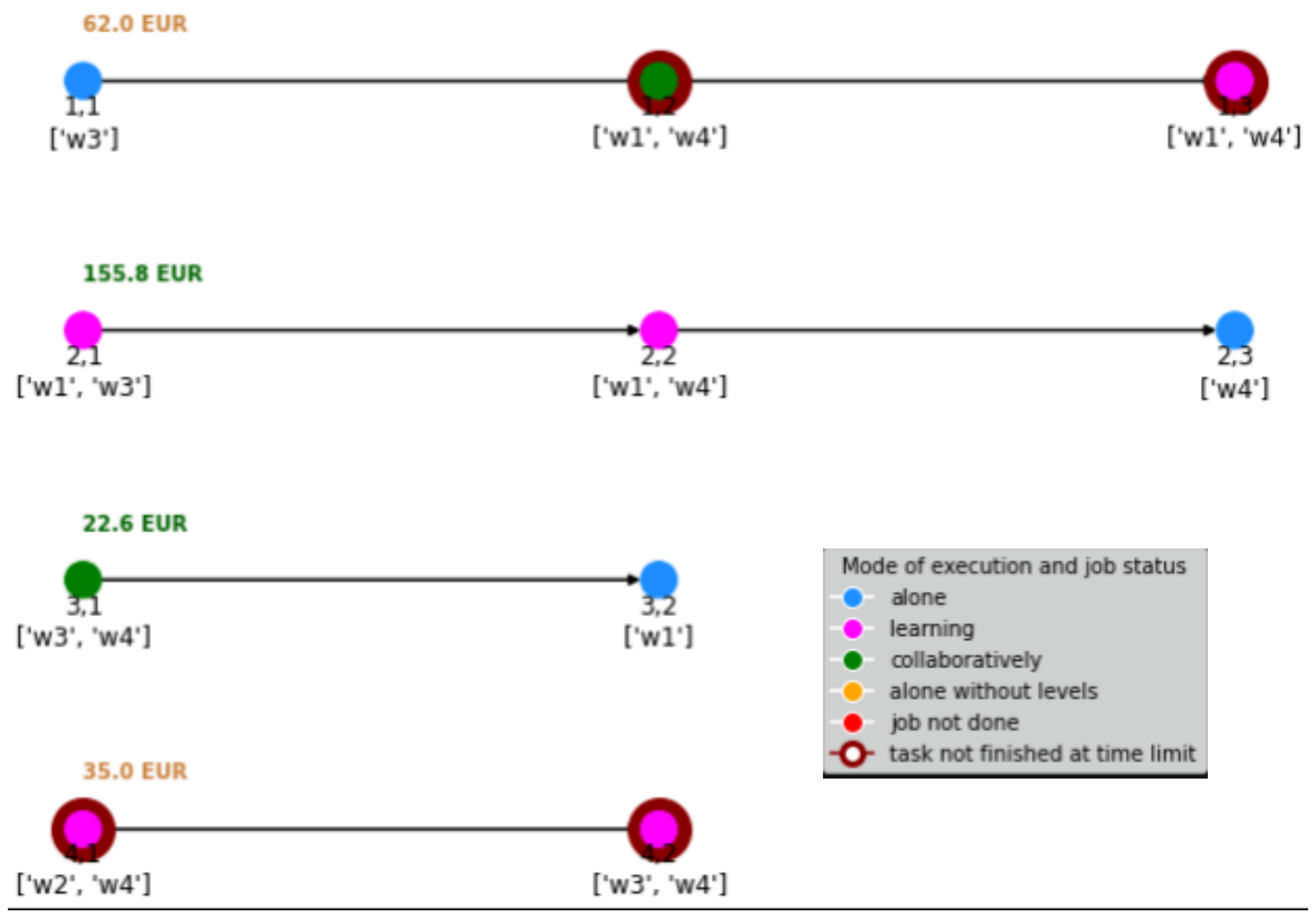


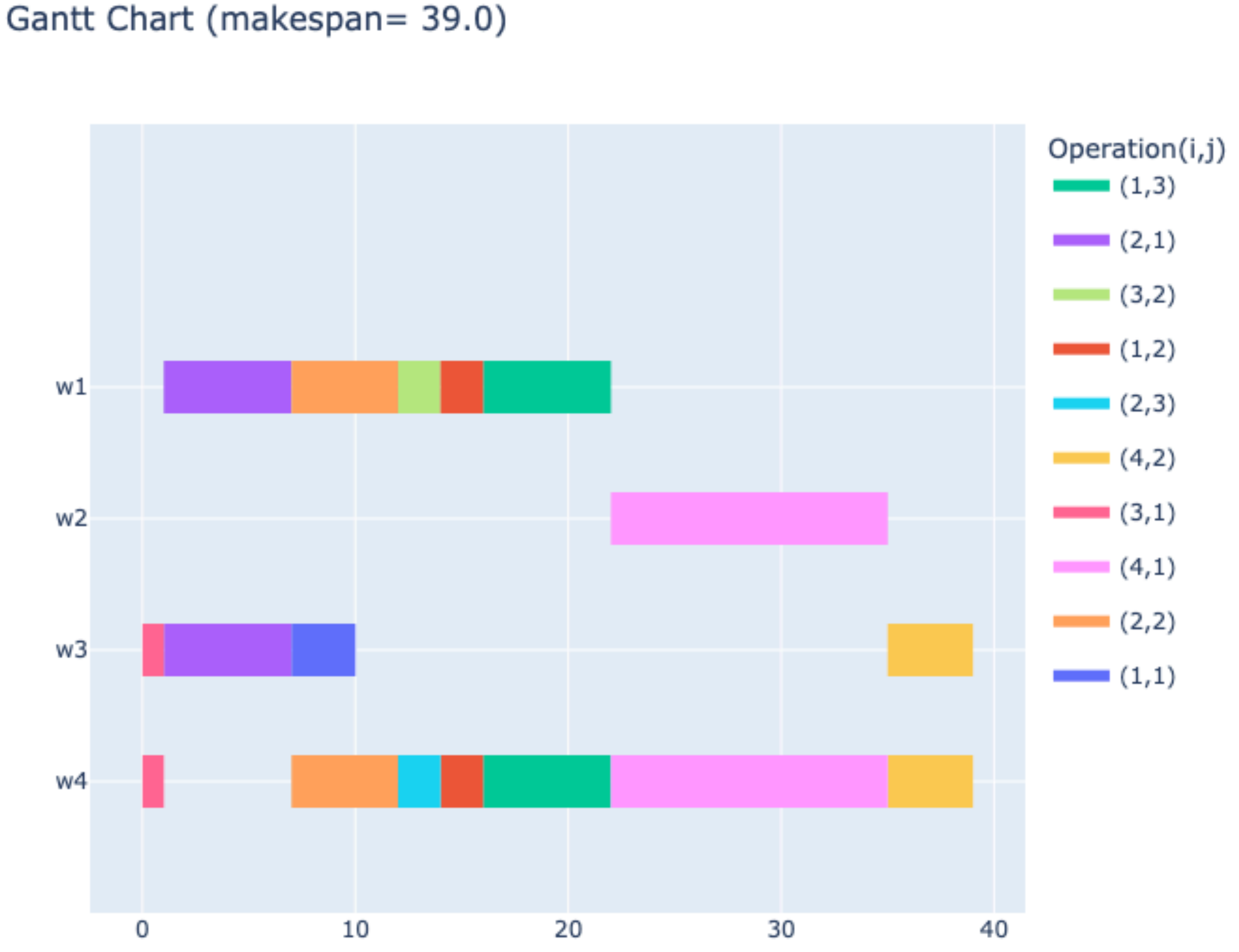


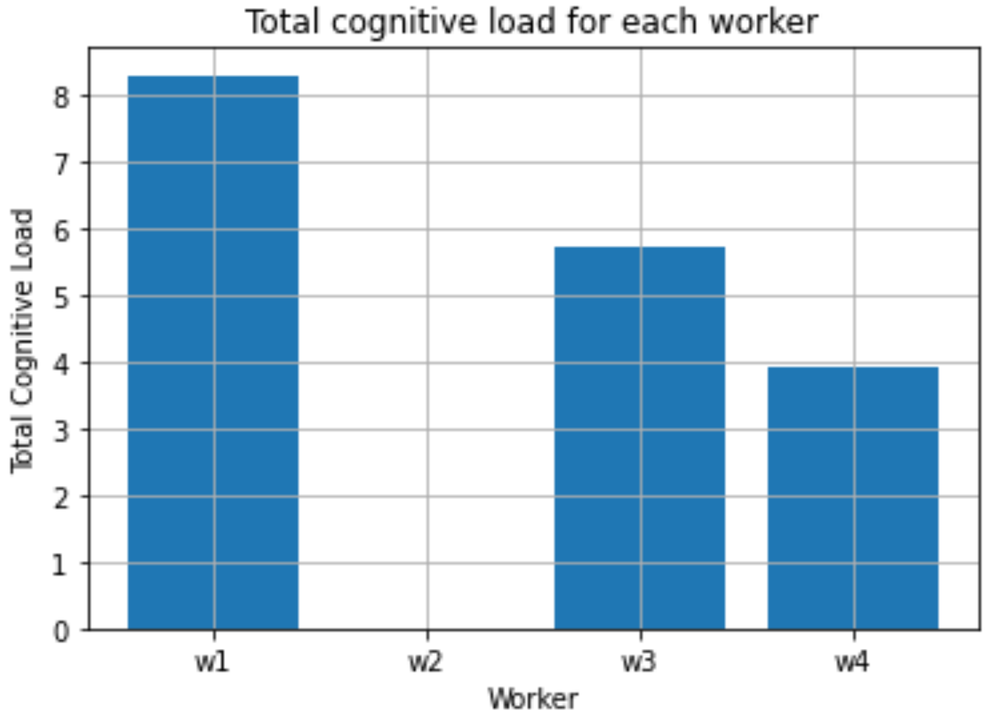


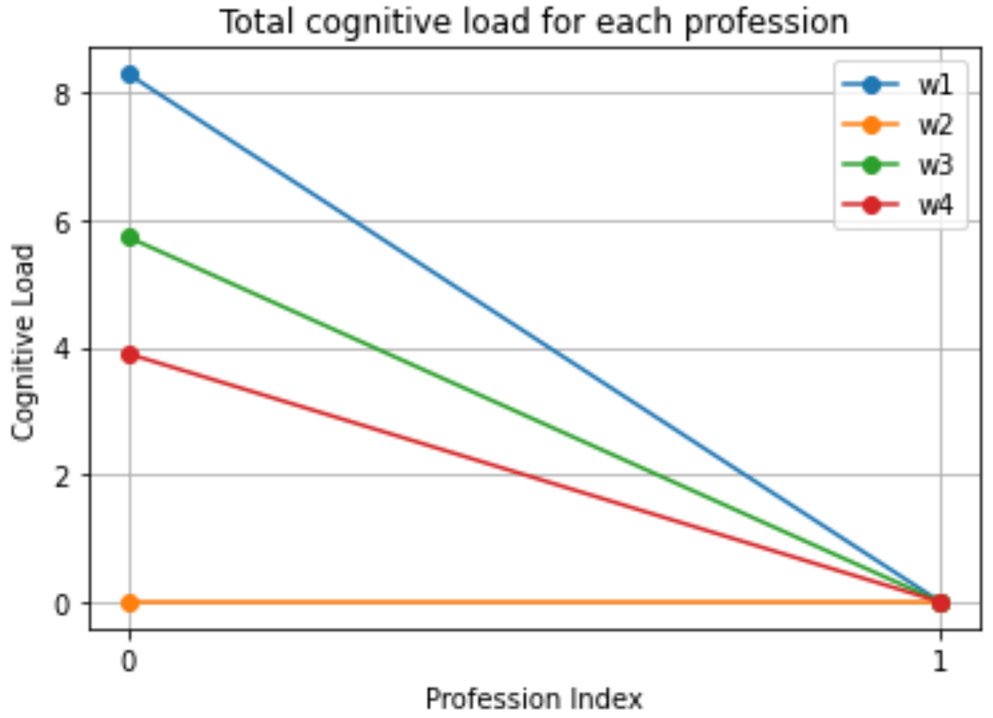
Local Search swap_op voisinage

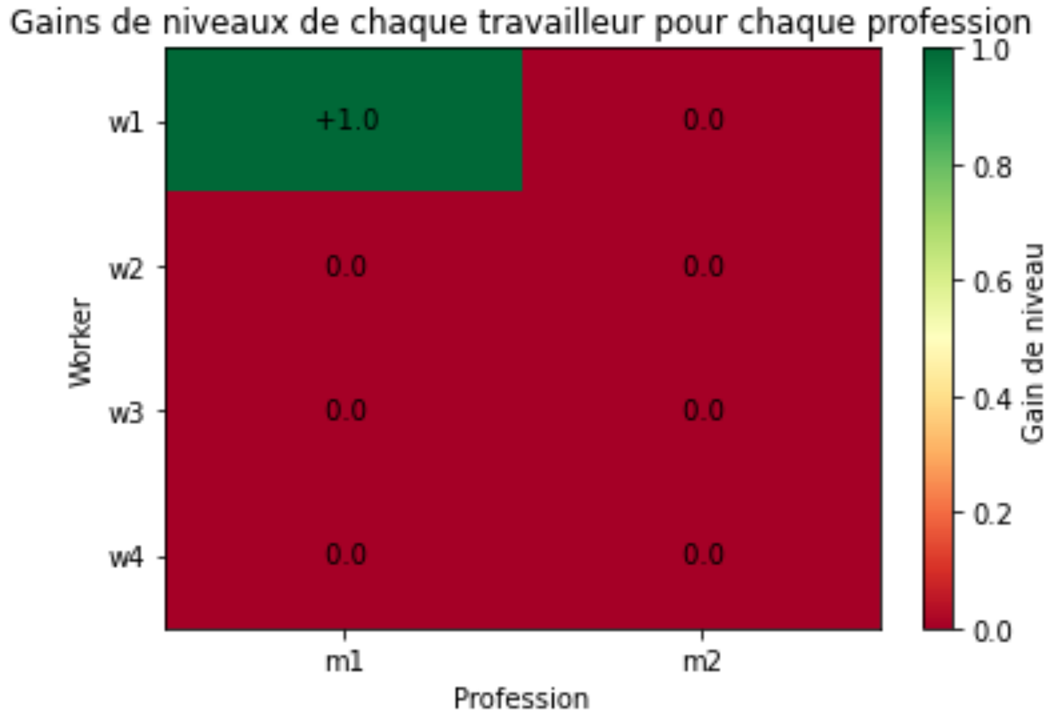
Génere n=10 voisins en permutant 2 opérations aléatoire

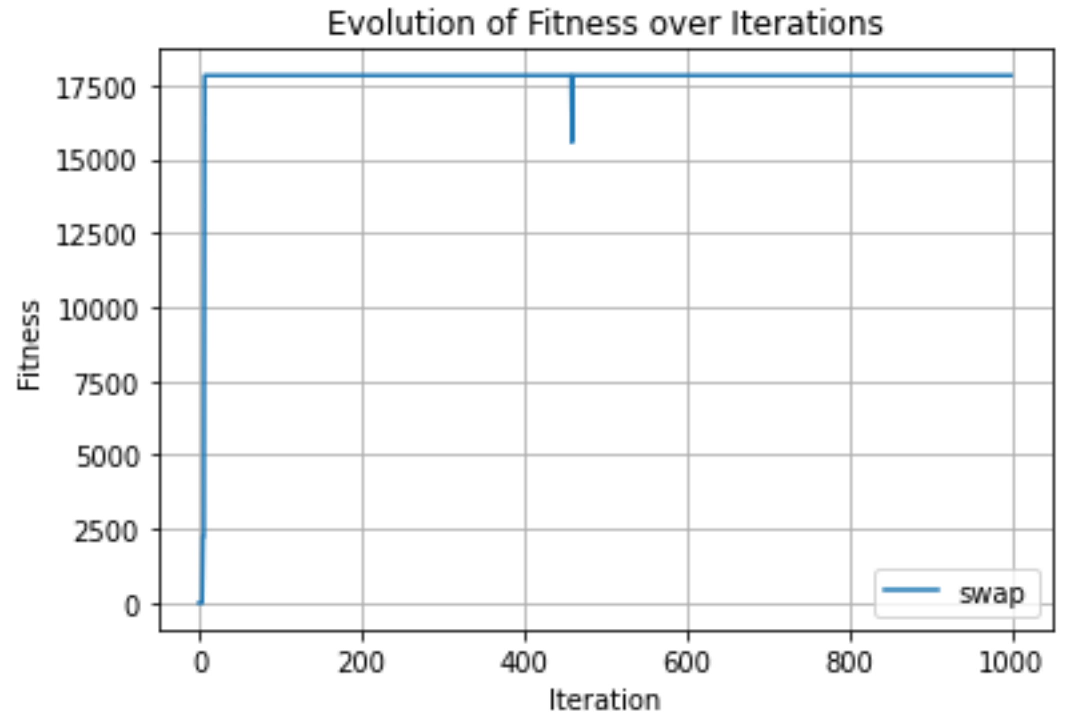






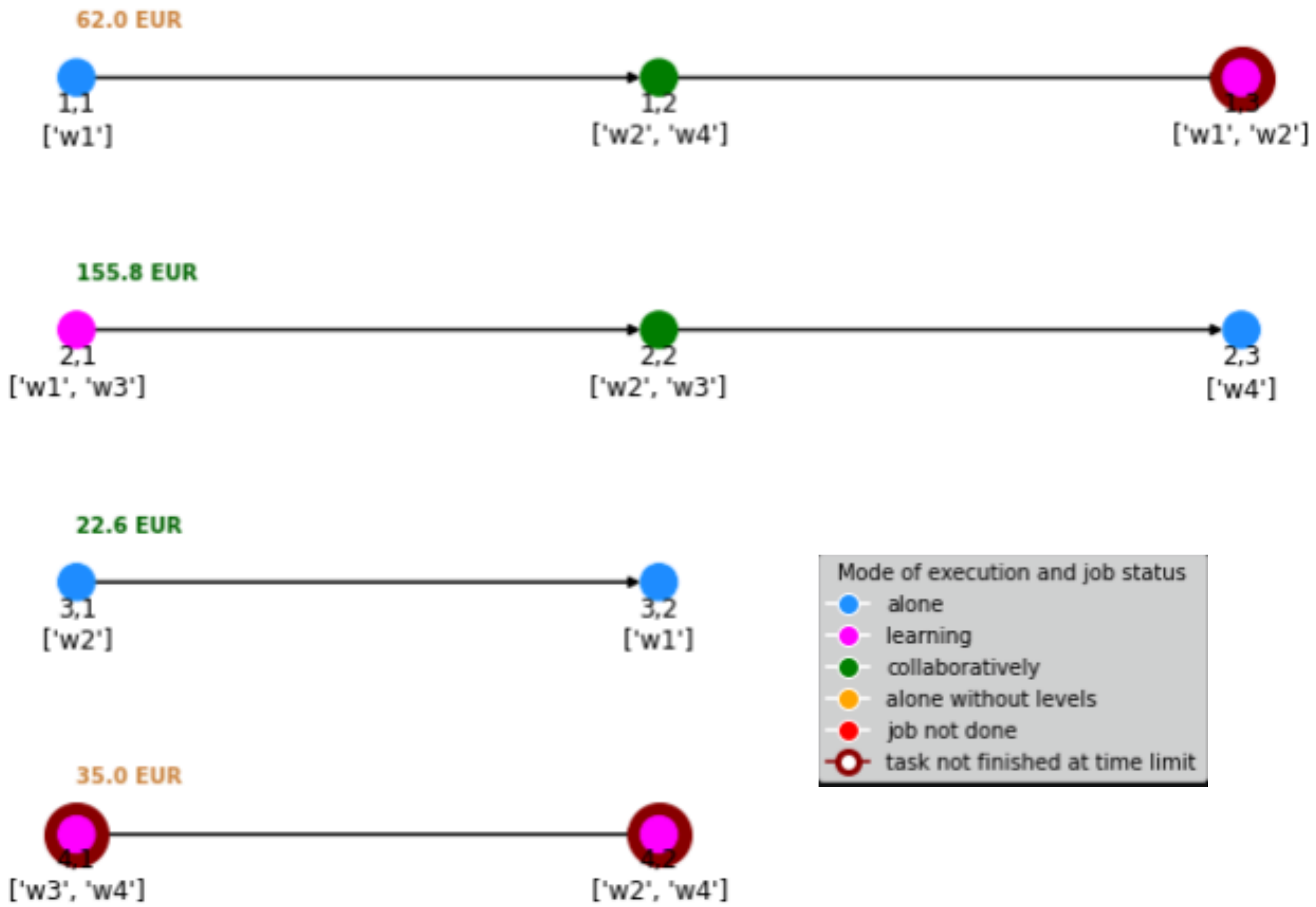


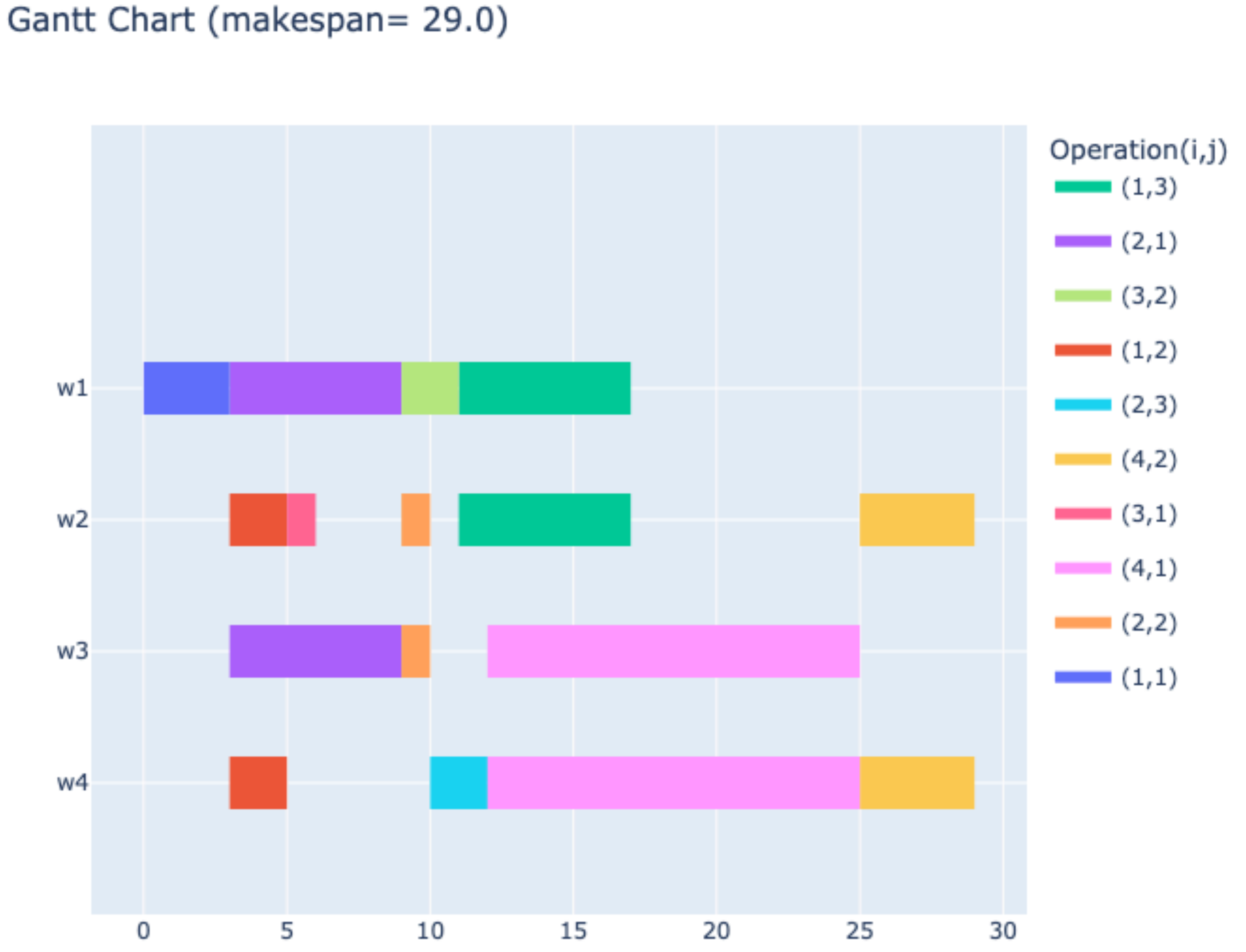


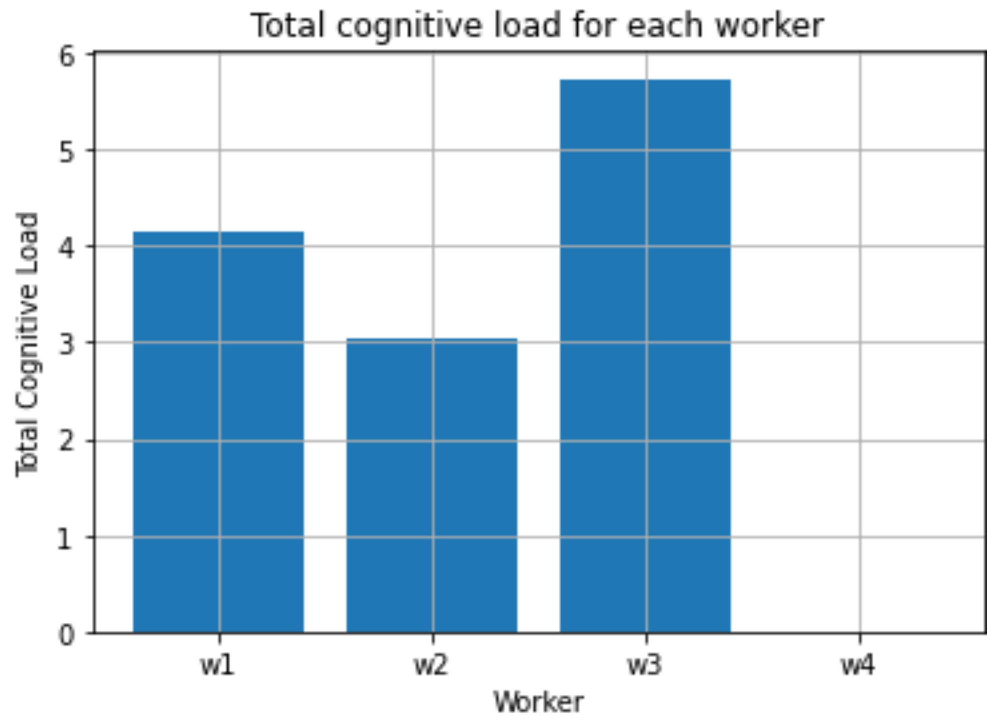


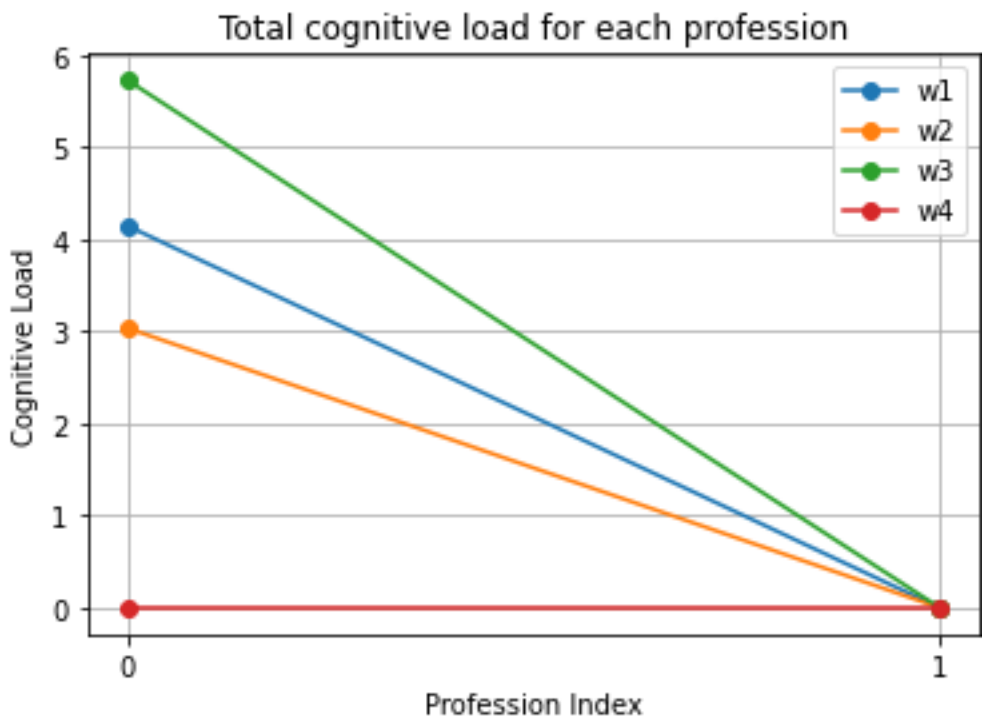
Local Search move_op voisinage

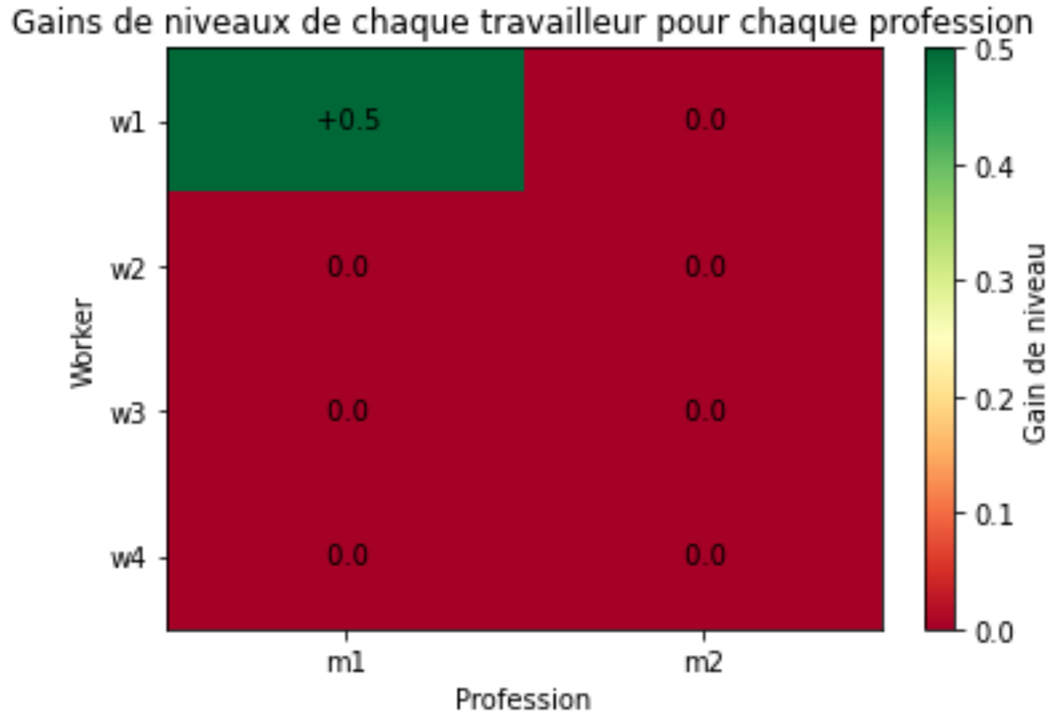
Chosit une opération aléatoire et la déplace à toutes les positions possibles

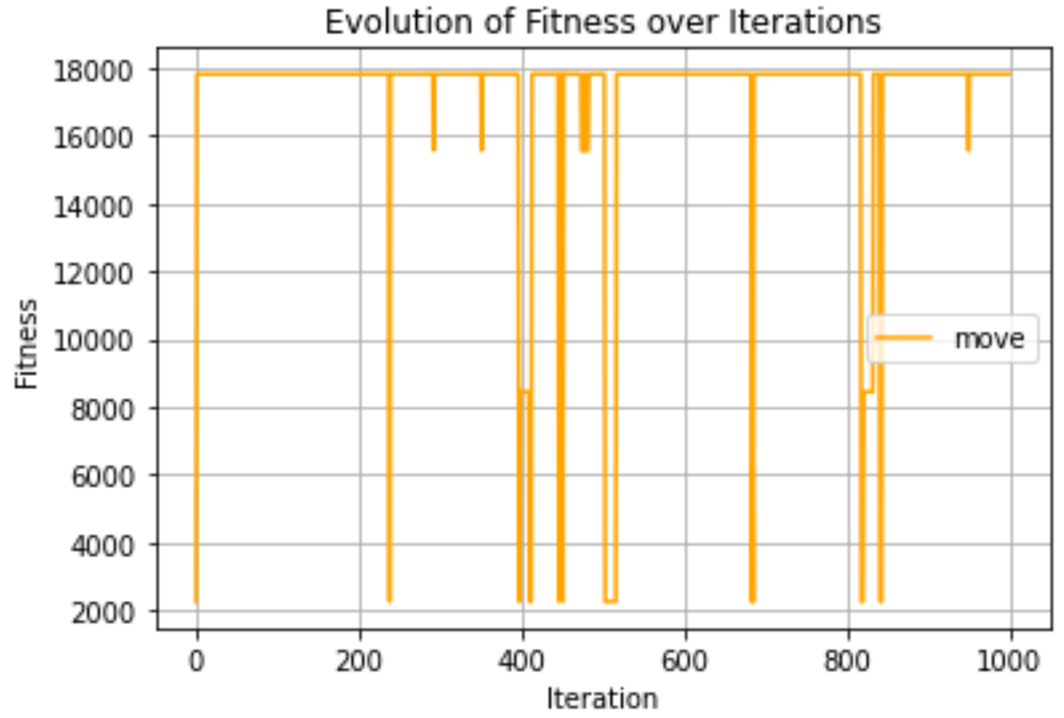






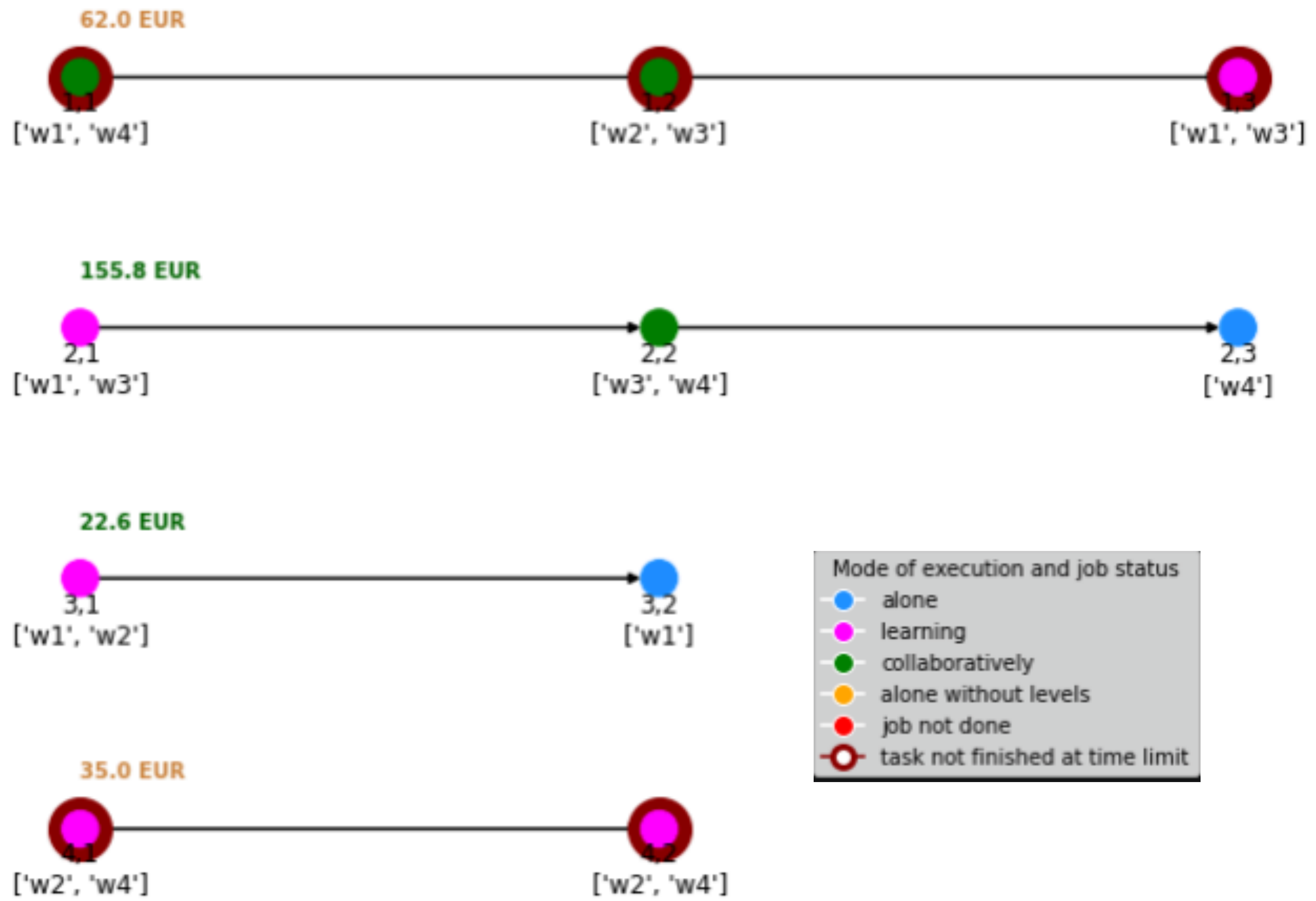


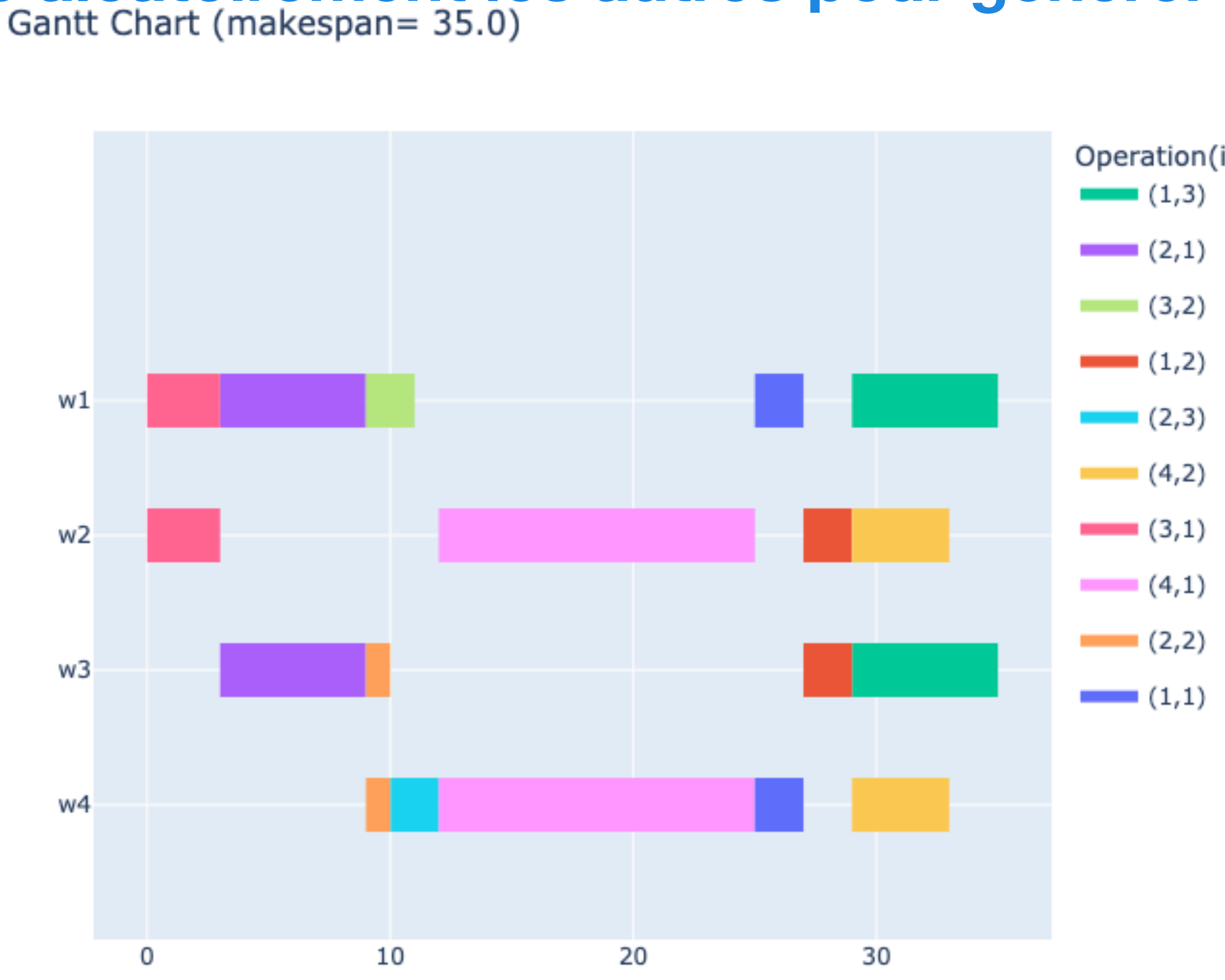


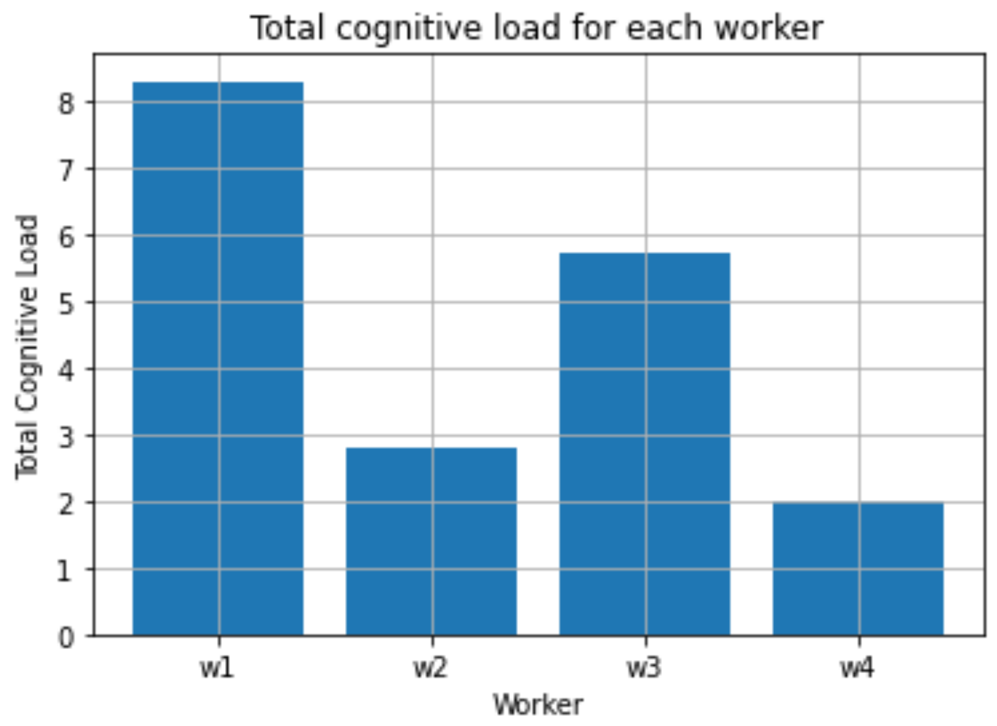


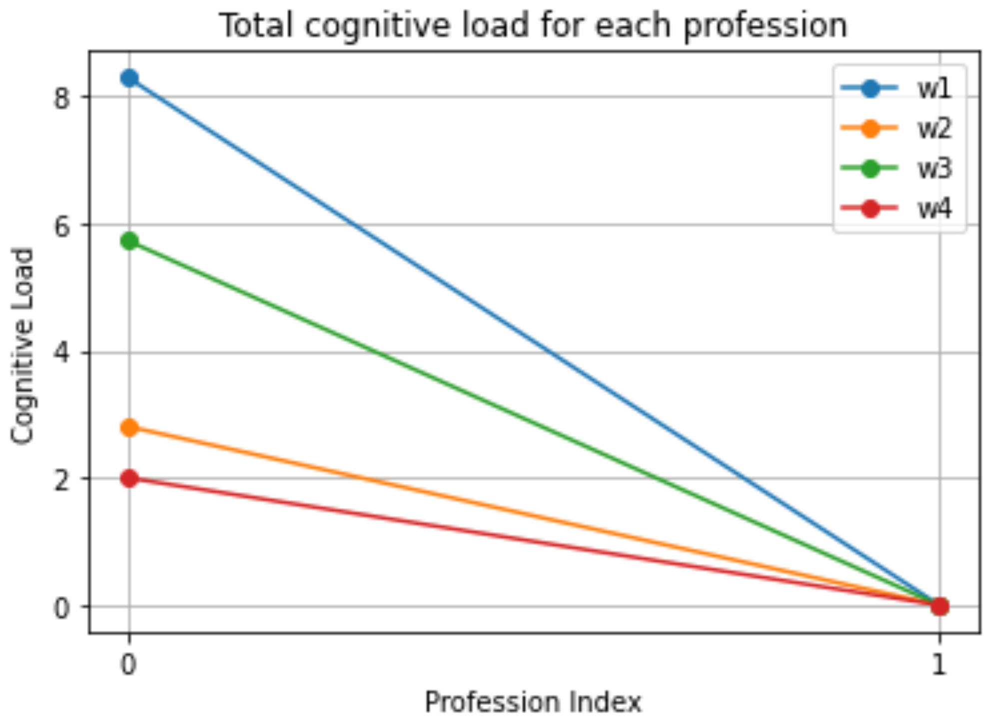
Local Search swap_fixed voisinage

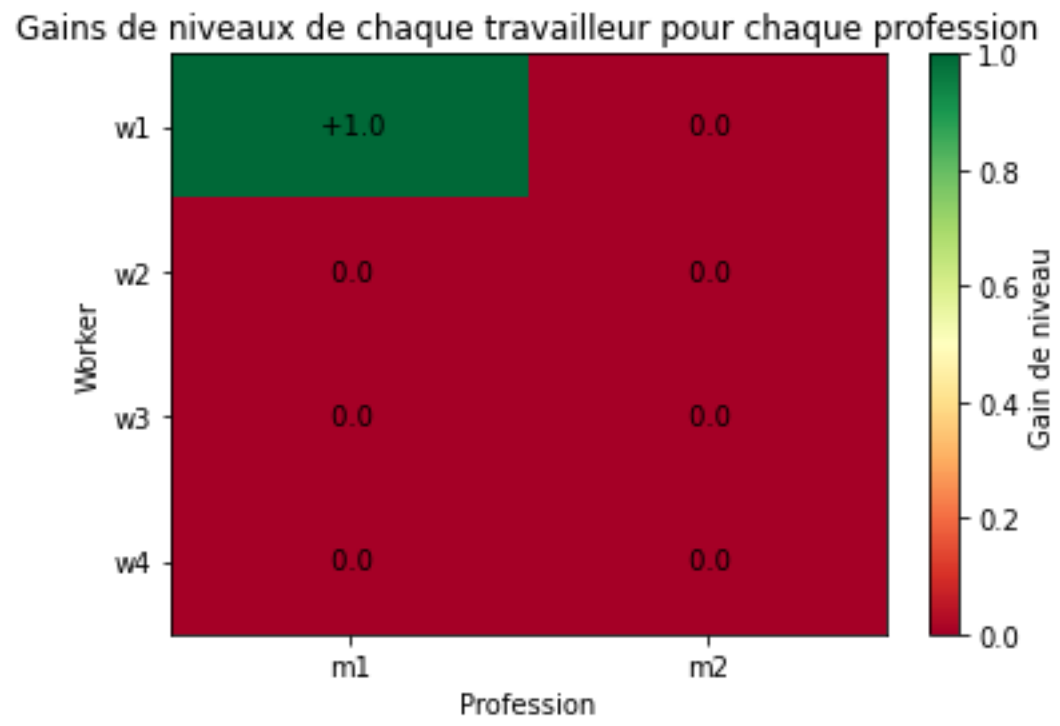
fixe k=10 opérations avec remise et permute aléatoirement les autres pour générer n=10 voisins

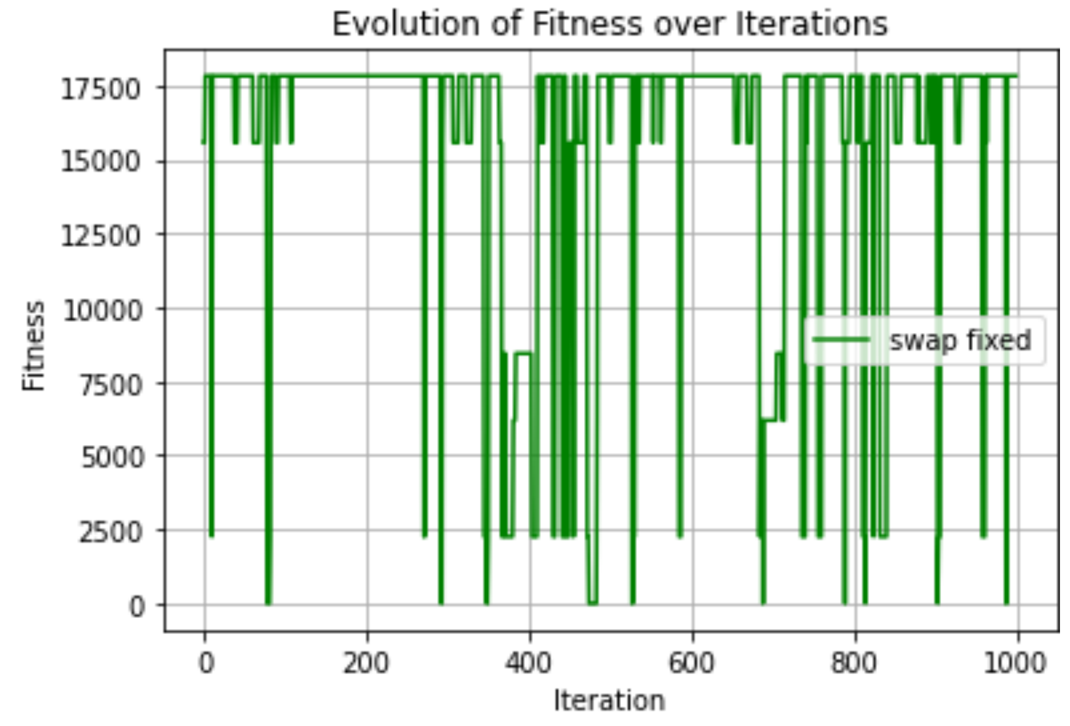






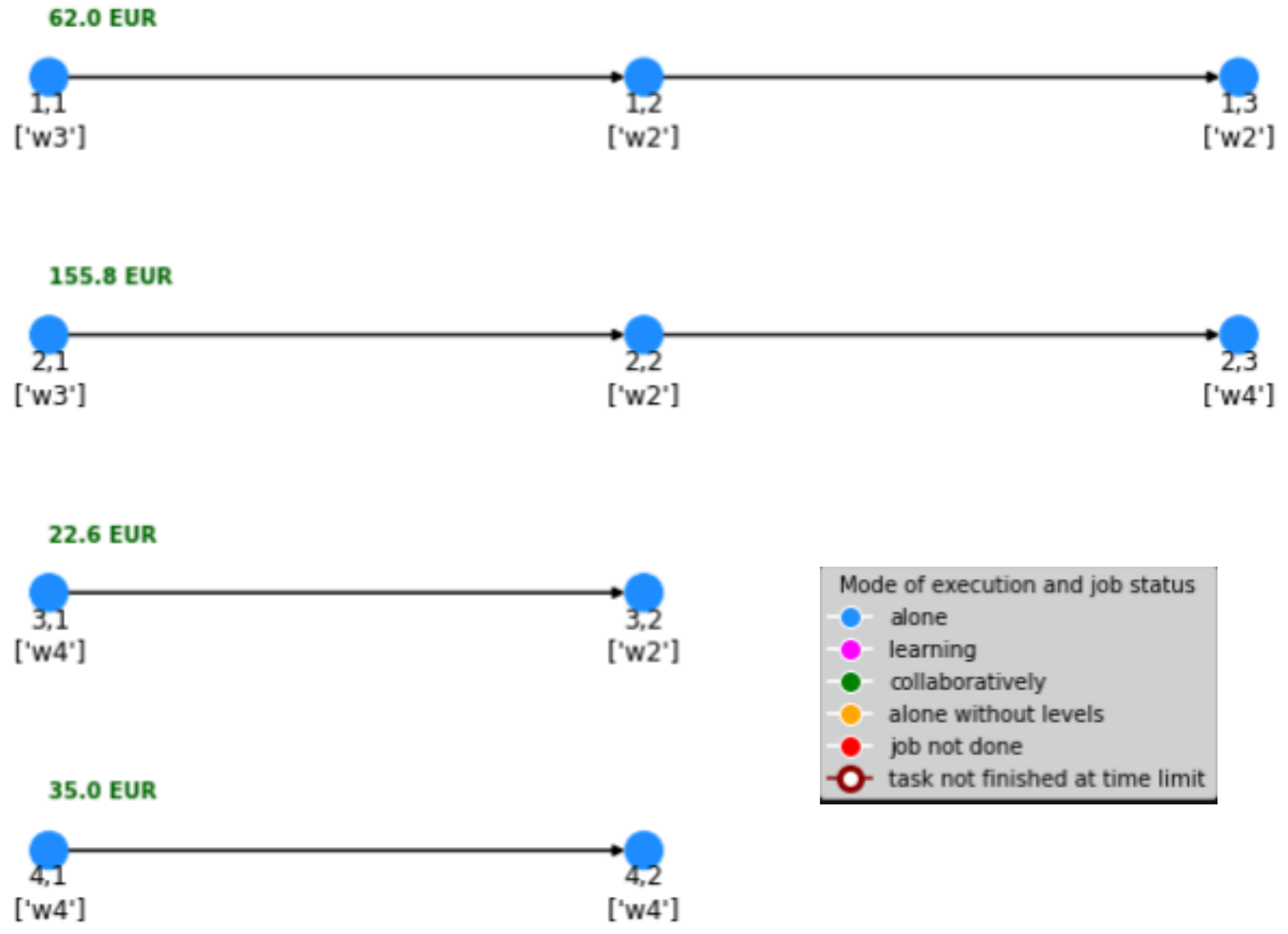


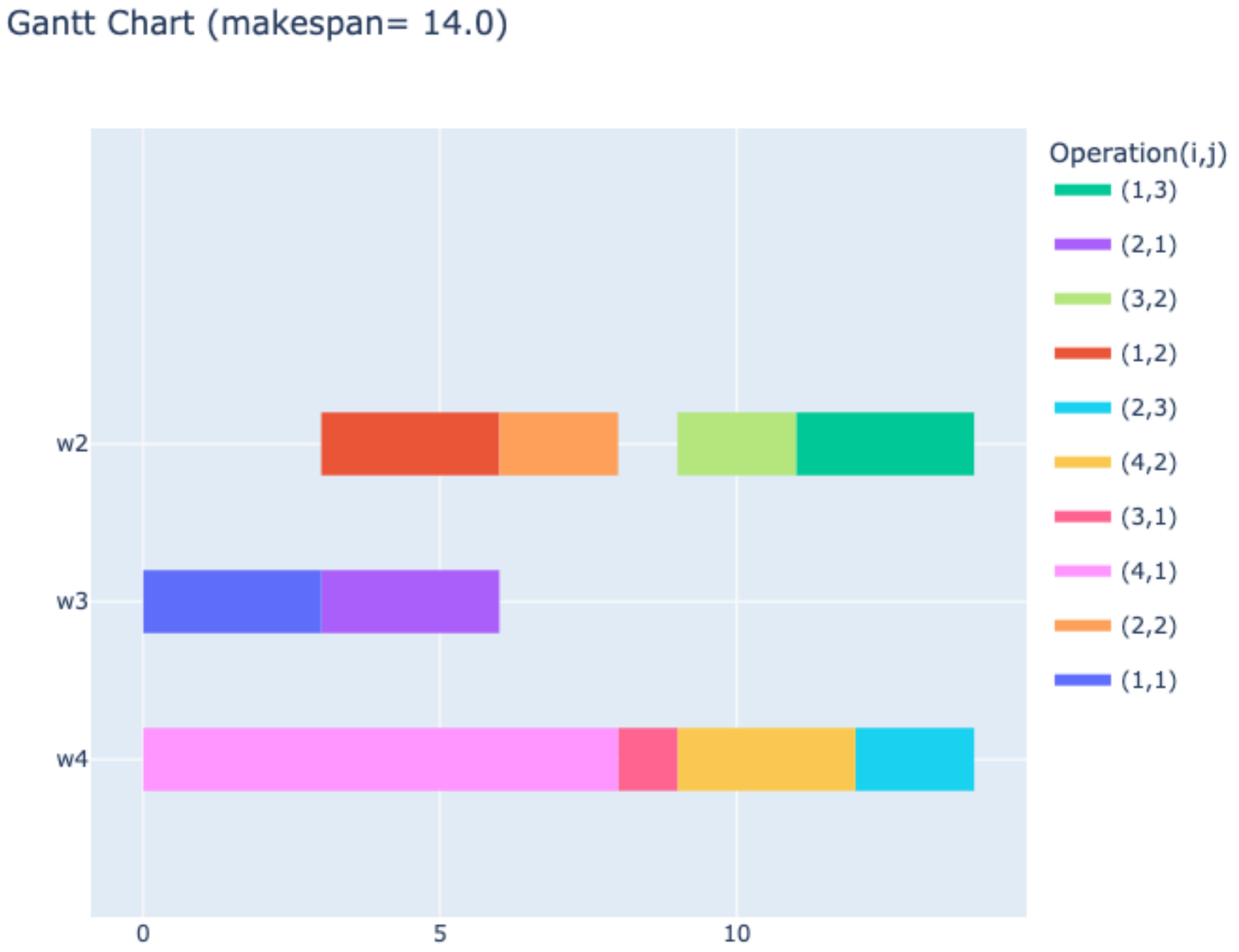


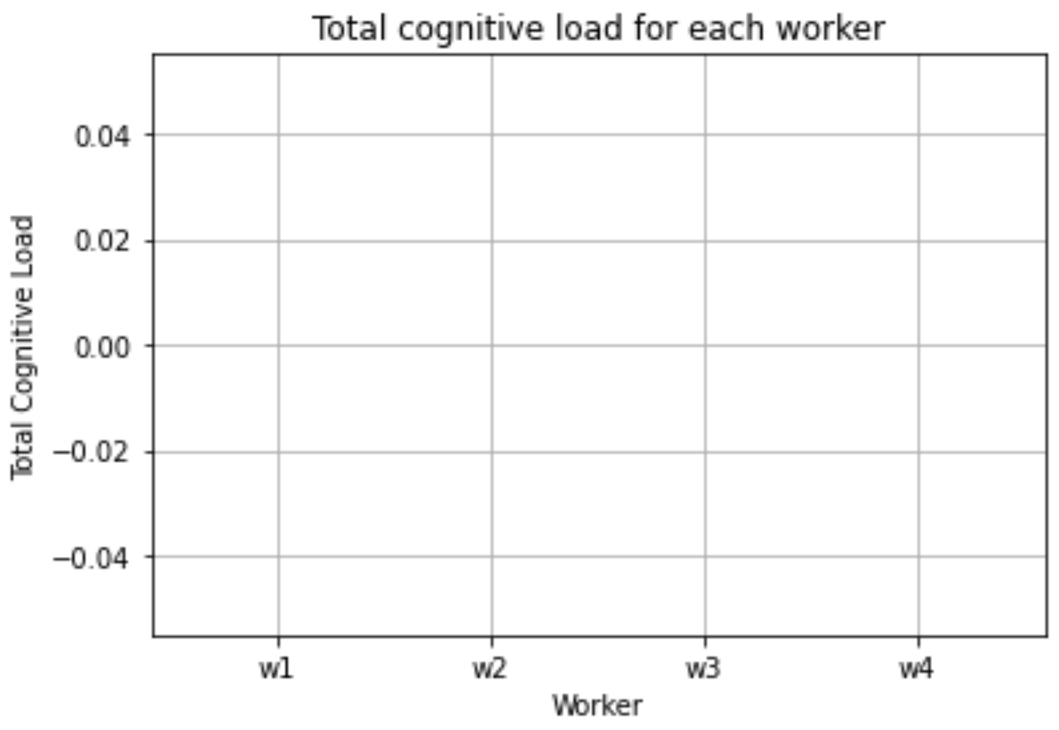


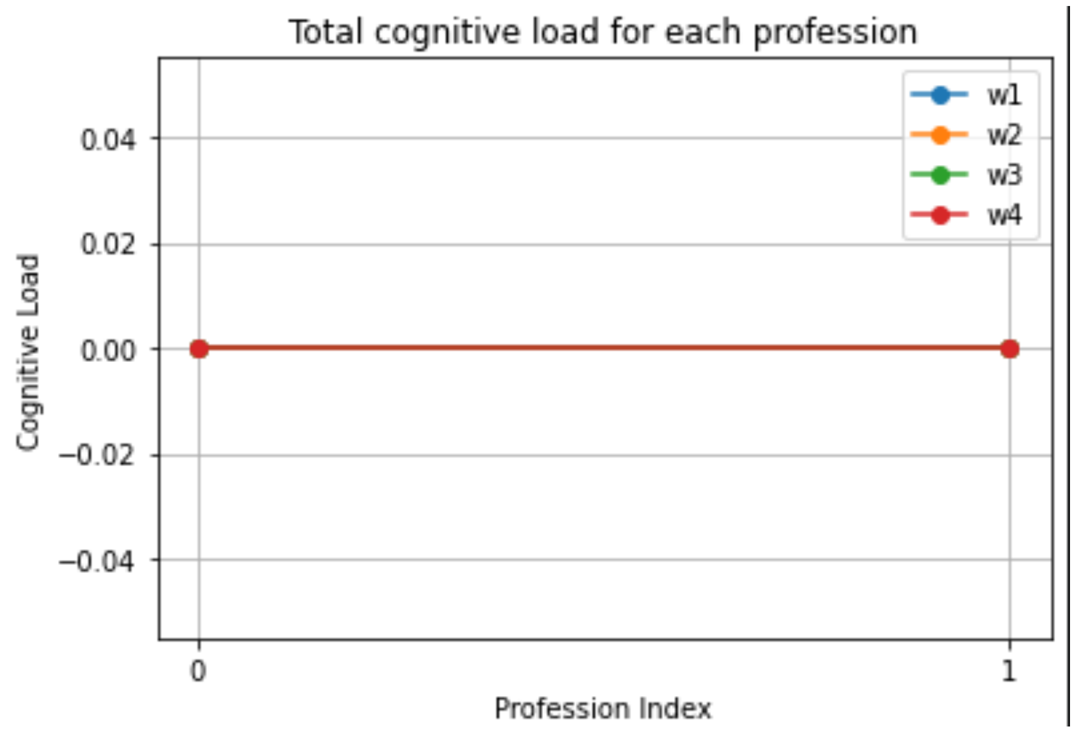
Local Search move_swac voisinage

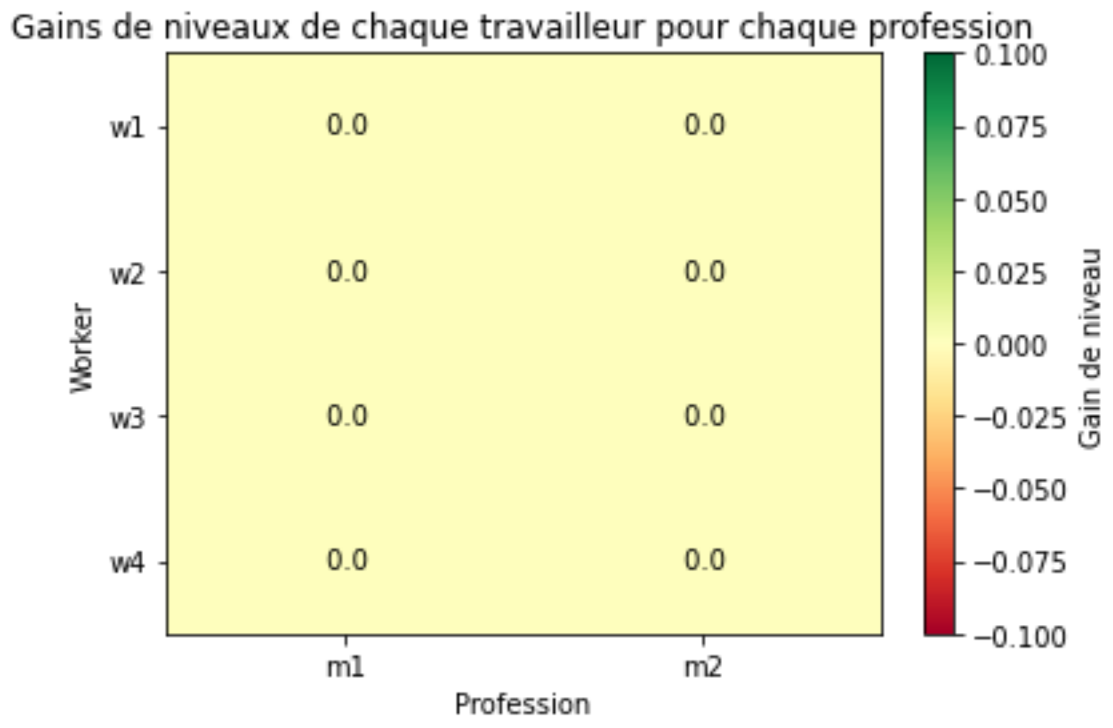
choisit une opération aléatoire et change le worker 2 de cette opération parmis tout ceux possible

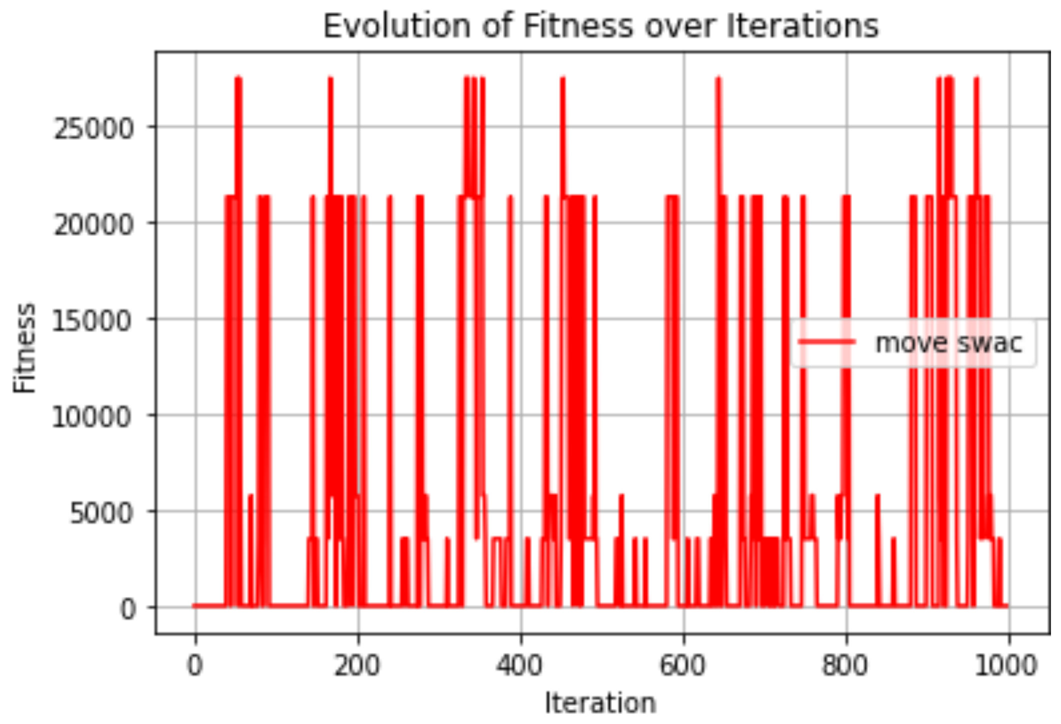






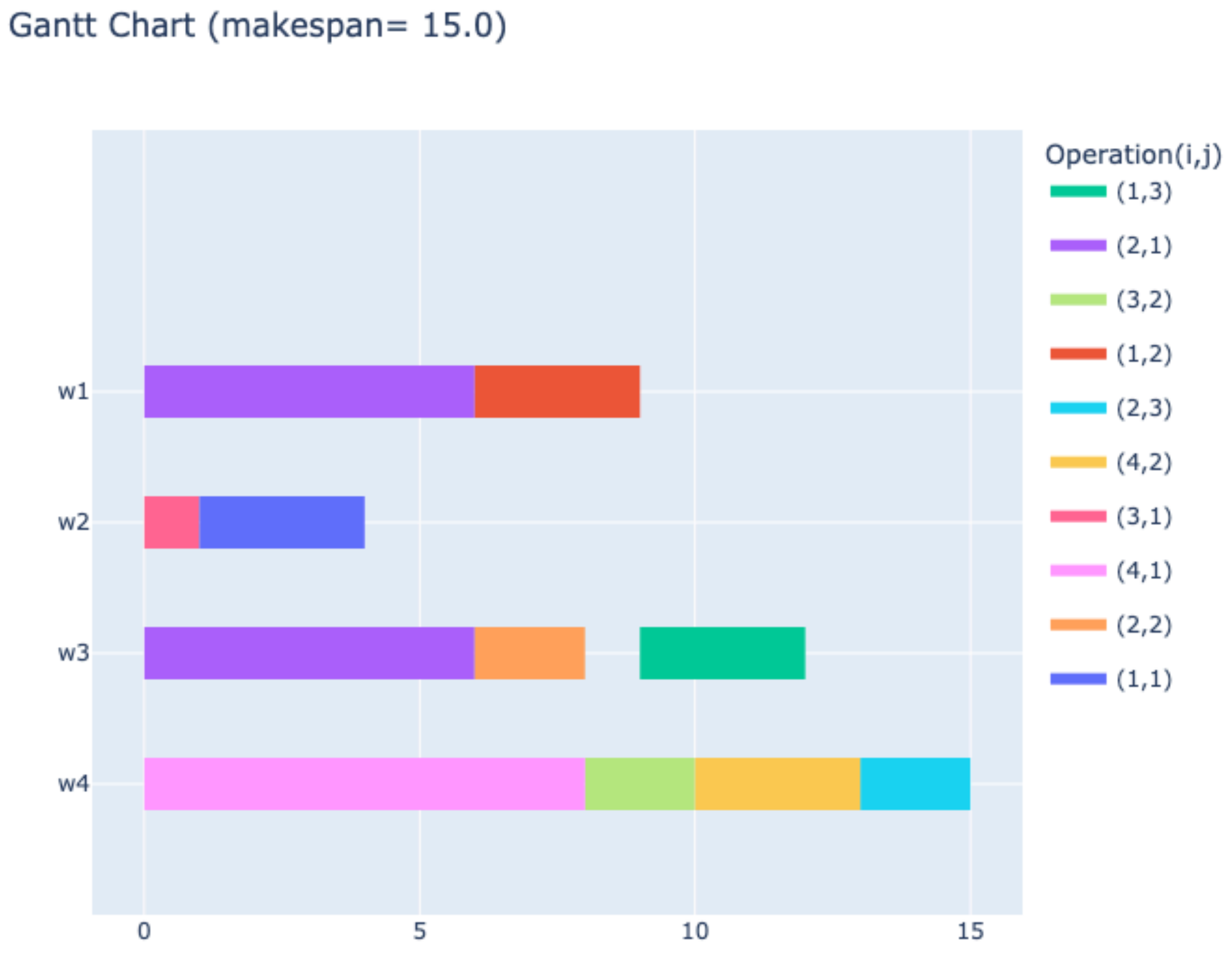


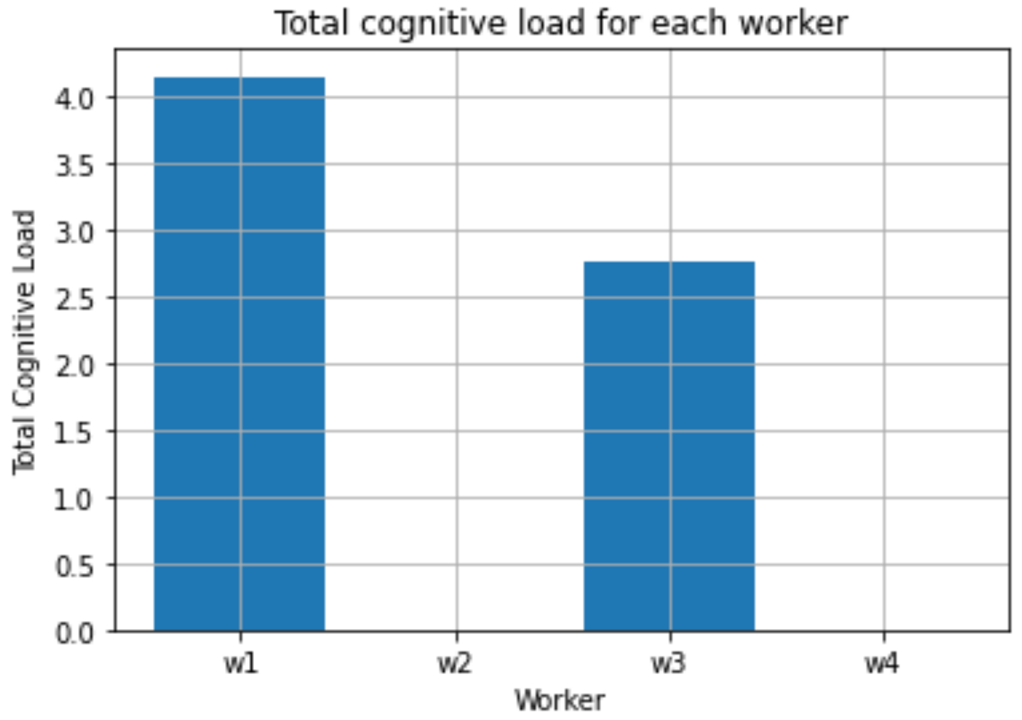


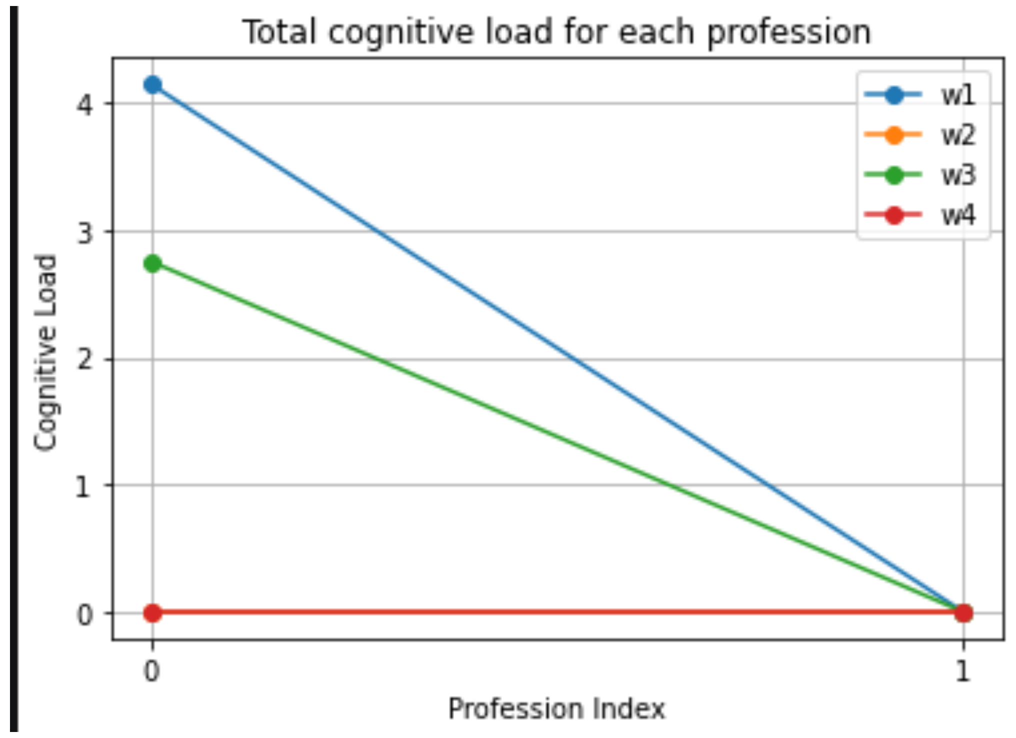


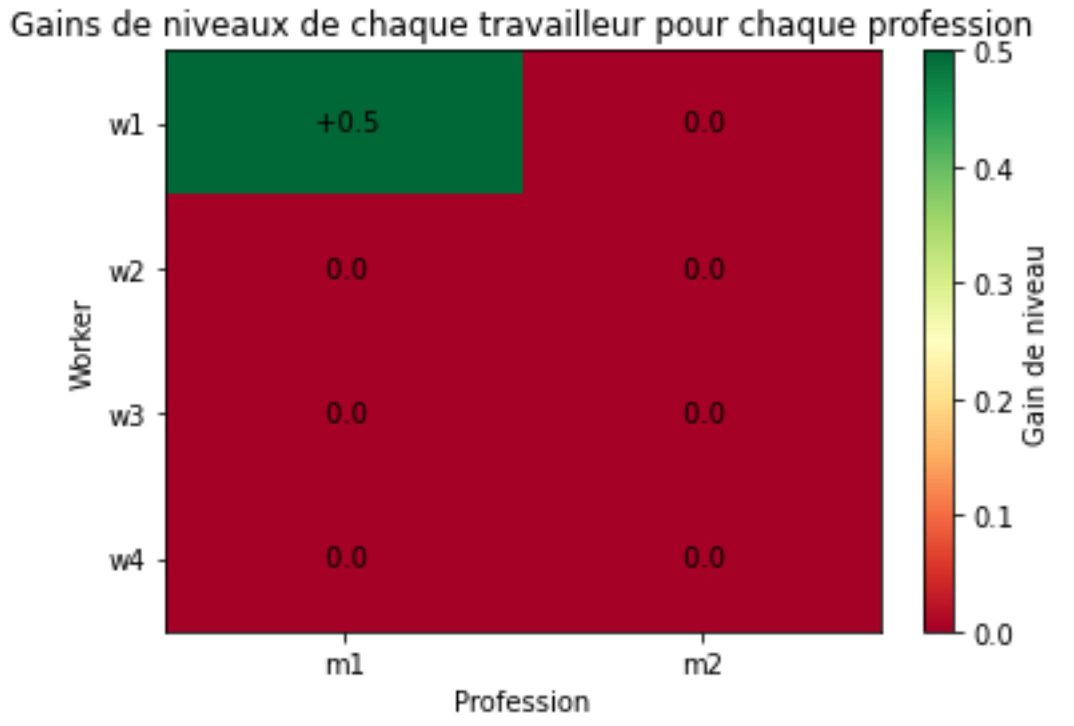
MILP Model

Trouve une solution optimal au problème

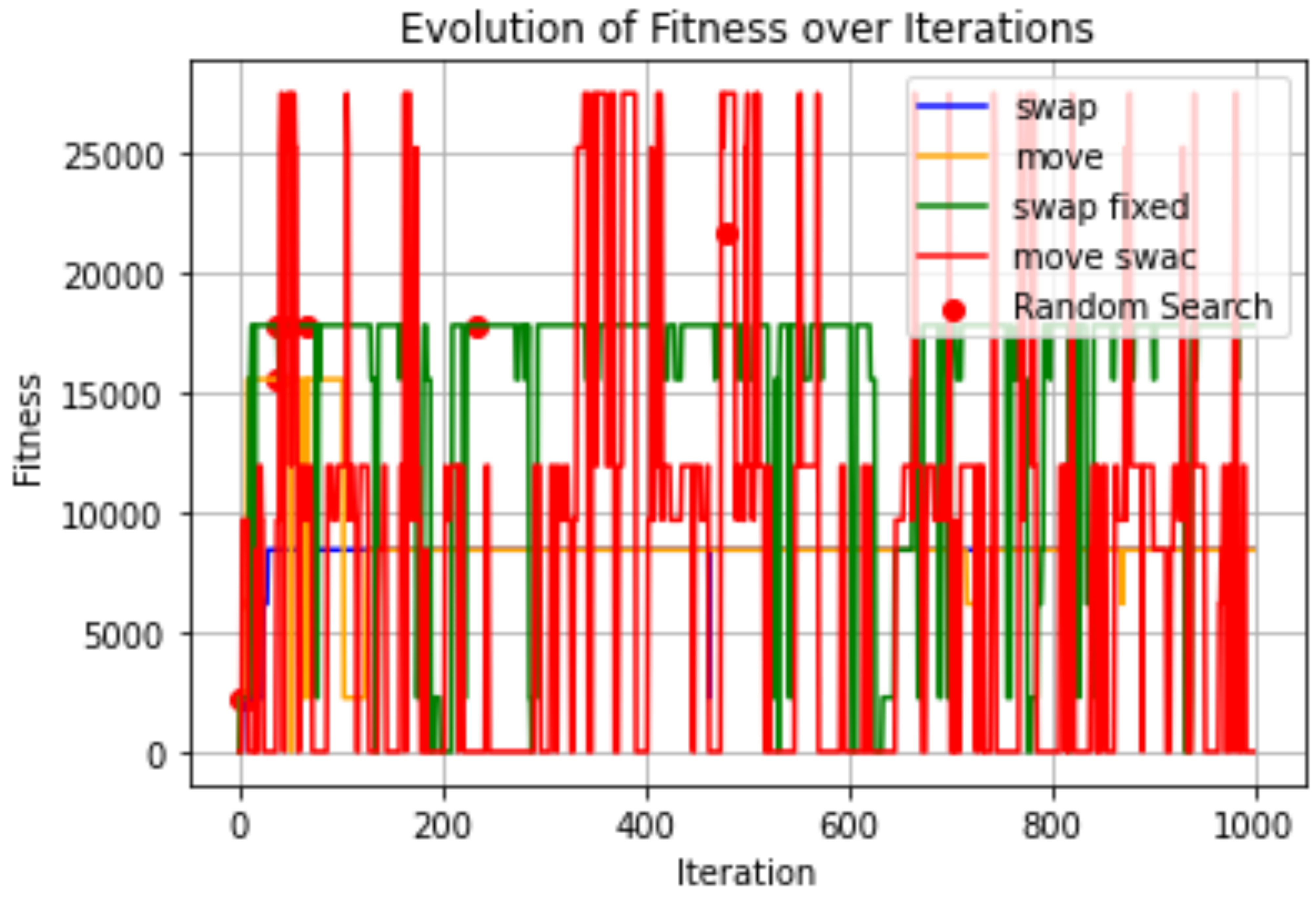




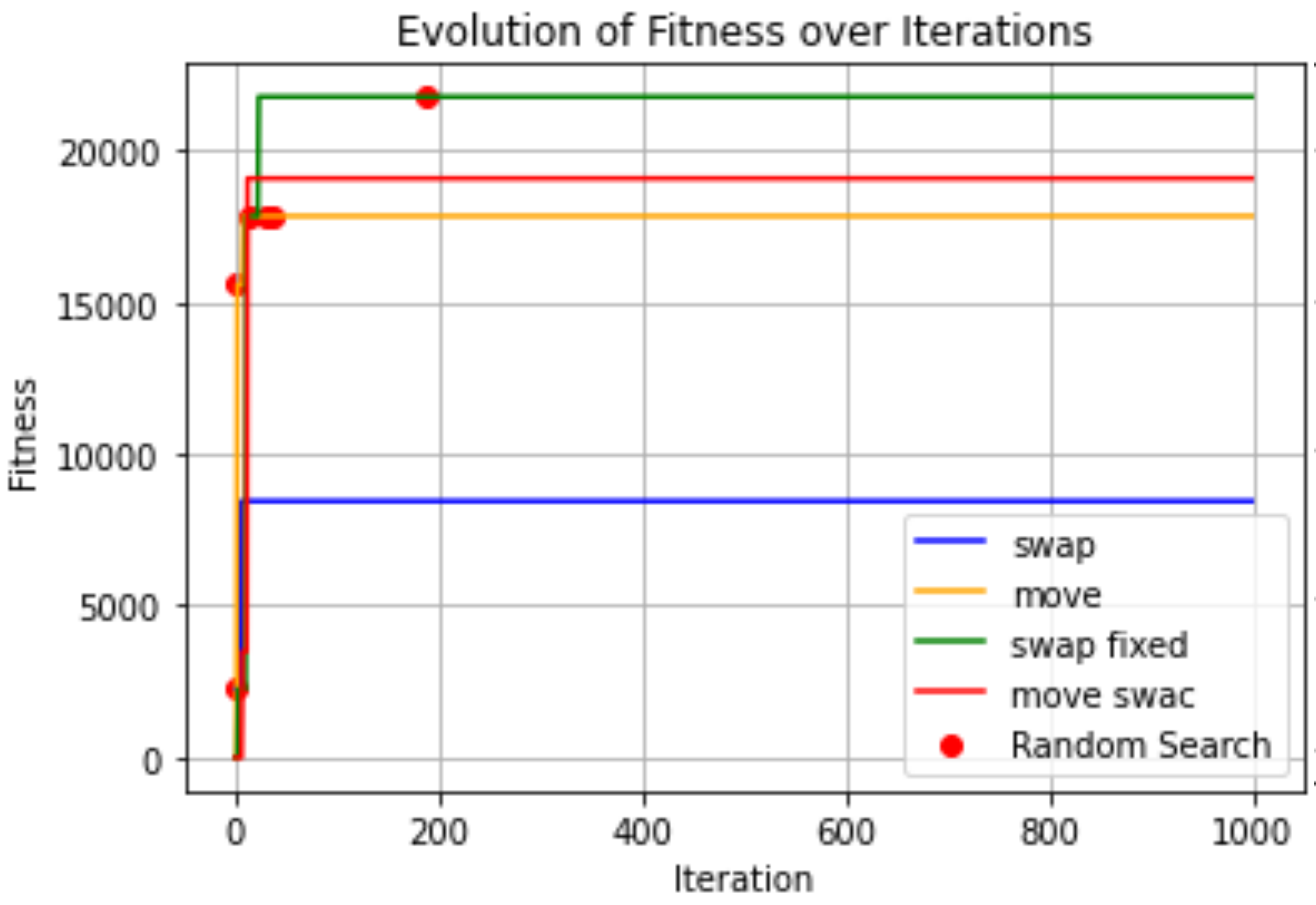


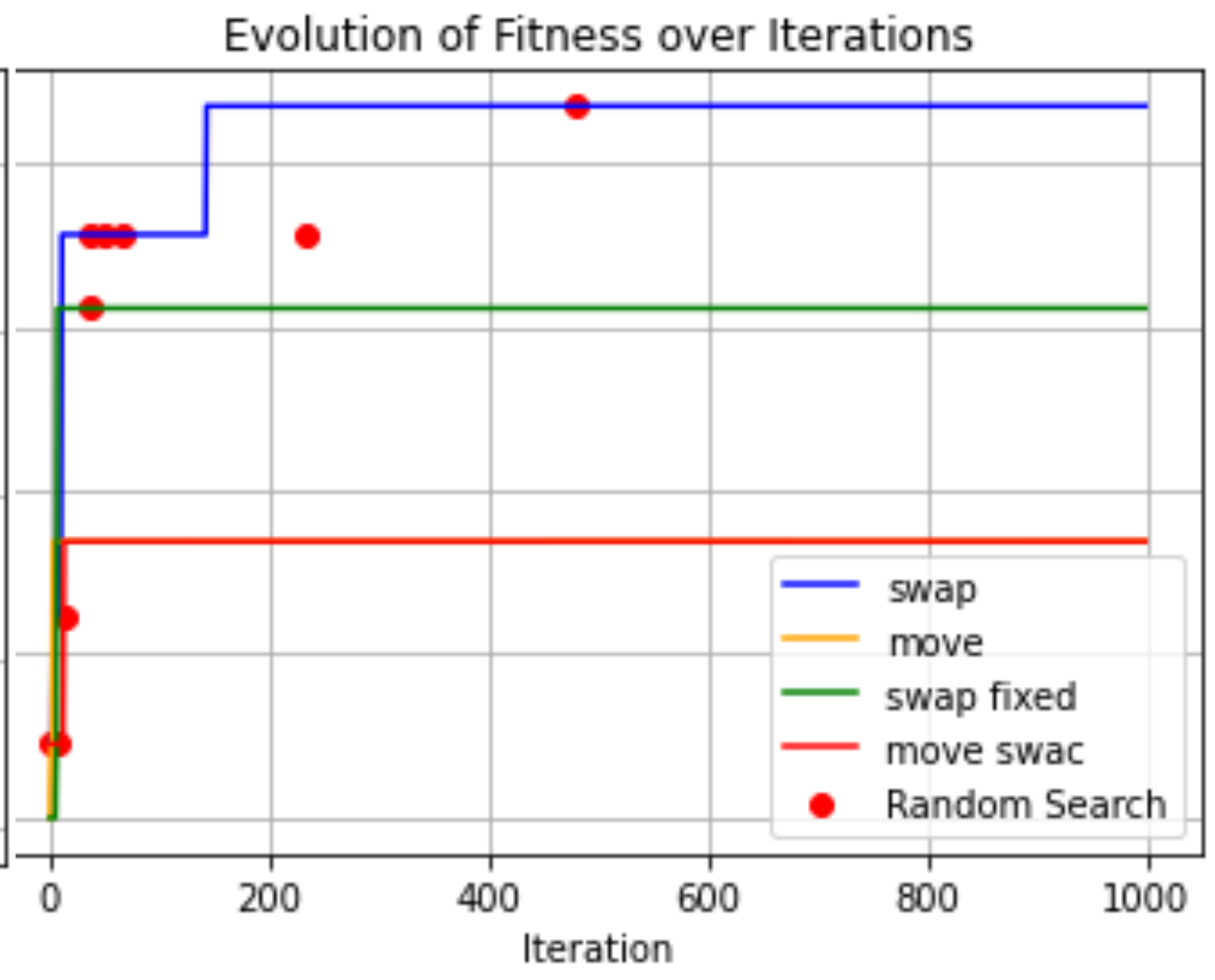


Candidat deviens le meilleur voisins même si moins bon que lui



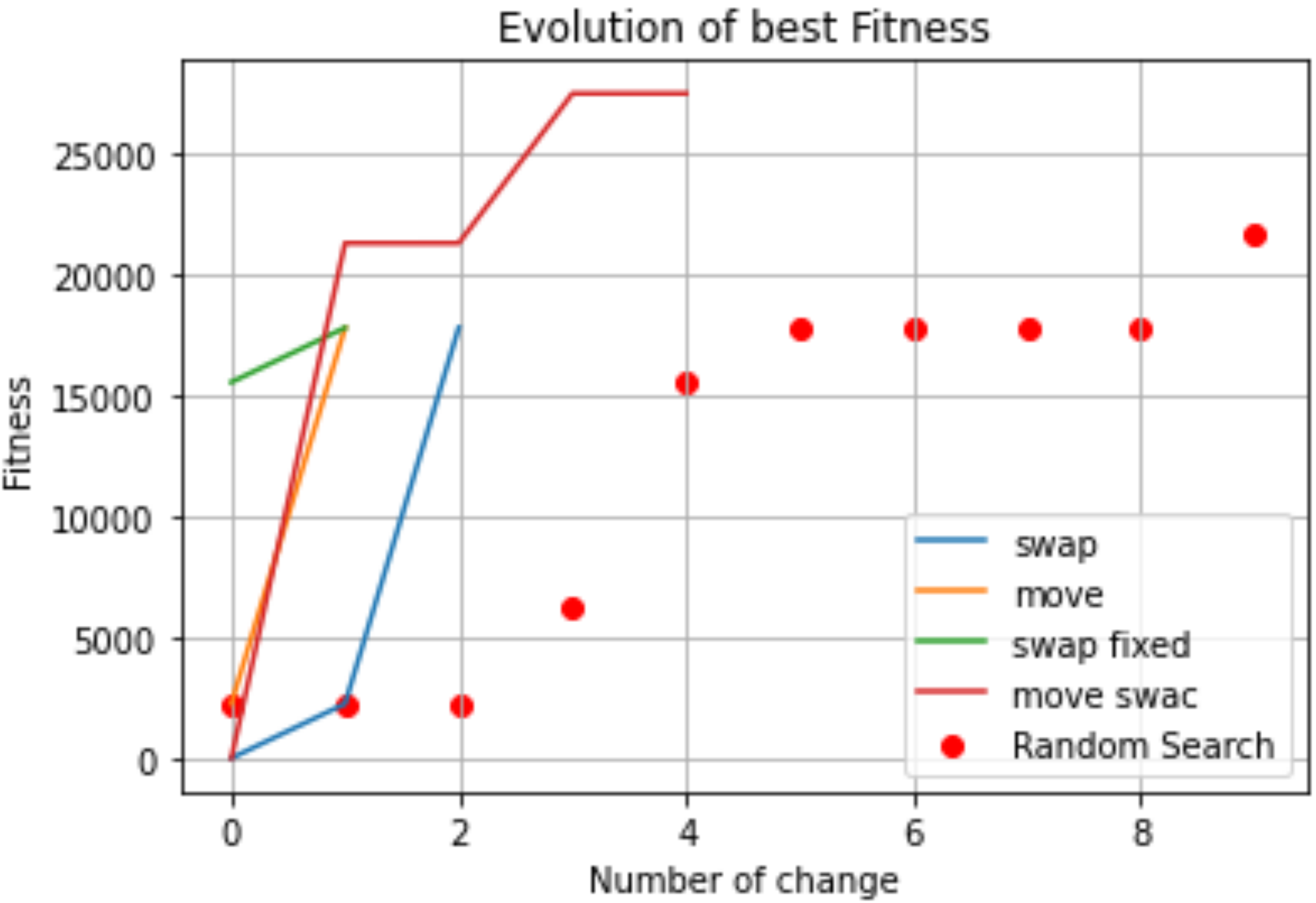
Candidat deviens le meilleur voisins seulement si meilleur que lui (2 seeds indépendantes)





Toujours pour la même instance on voit que les résultats varient énormément

Récapitulatif



	Profit	Skills	Load	Agg
Rand_Search	217.72	1	20.90	~21 762
swap_op	178.36	1	17.92	~17 828
move_op	178.36	0.5	12.91	~17 828
swap_fixed	178.36	1	18.83	~17 827
move_swac	275.29	0	0	~27 529
MILP	275.29	0.5	6.90	~27 527

Evolution de la fitness selon le nombre de fois que l'algo à rencontré un meilleure candidat

Problème : Solution du MILP dominé par move_swac